

Basic Financial Model

Jakarta, 22 May 2026

What is Financial Model ?

What is a Financial Model

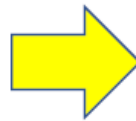
A Financial Model is a tool that can be used **to forecast a picture of financial performance of a business** (a project) based on certain assumptions.

A Financial Model can be used for an analysis of a business (a project) such as **investment judgement, setting sales prices**, or **business risk evaluation**.

Example of a Financial Model

Assumptions

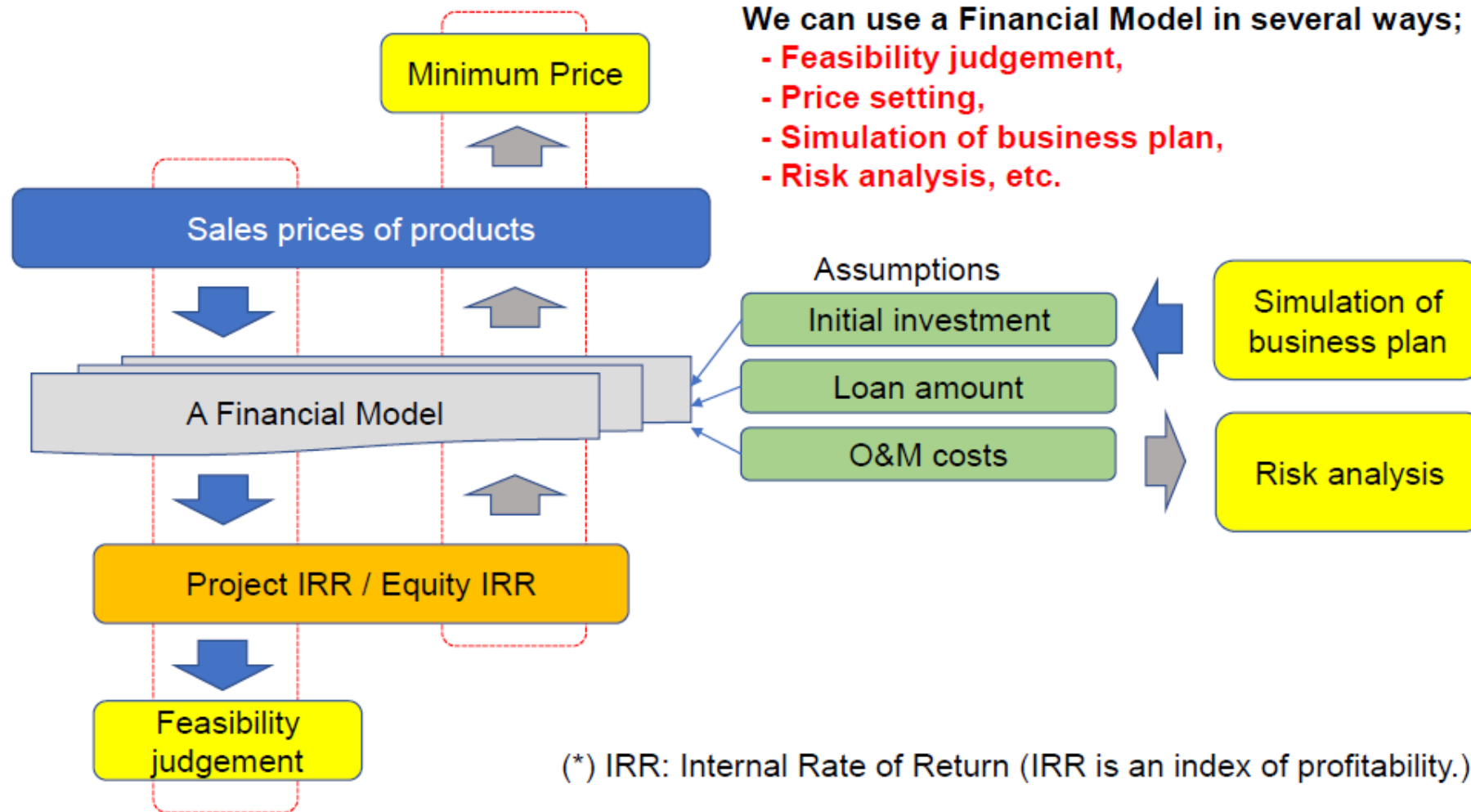
	A	B	C	D	E	F	G
1							
2				Sample case			
3							
4				Output (gross)		50 MW	
7				Salable energy		370 GWh	
8				Sales price of energy		10.0 USct/kWh	
9				Annual sales		37.0 M\$	
10							
11				Construction cost		150 M\$	
12				Construction period		3 years	
13				Construction cost disbursement			
14				-3 year		40 M\$	
15				-2 year		60 M\$	
16				-1 year		50 M\$	
17				Total		150 M\$	
18							
19				Finance			
20				Equity		60 M\$	
21				Borrowing		90 M\$	
22				-2 year Borrowing		50 M\$	
23				-1 year Borrowing		40 M\$	
24				Loan period		5 years	
25				Interest rate		6 % a.a.	
26				Principal equal repayment			
27				Expected Return on Equity (ROE)		10 %	
28				WACC		7.6 %	
29							
30				Operation period		10 years	
31				Annual operating cost		7.0 M\$ annual	
32				Depreciation period		10 years	
33				Depreciation method		Straight-line	
34							
35				Tax rate		30 %	



Forecast of Financial Performance

A	B	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
37																			
38	Item		Unit			-3	-2	-1	1	2	3	4	5	6	7	8	9	10	Total
39	STEP 1.1 CAPEX schedule																		
40	CAPEX expenditure	(0)	MS			-40.0	-60.0	-50.0											-150.0
41	Equity	(0)	MS			40.0	10.0	10.0											60.0
42	Borrowings	(0)	MS				50.0	40.0											90.0
43																			
44	STEP 1.2 Annual sales																		
45	Energy sales amount	(1)	GWh						370.0	370.0	370.0	370.0	370.0	370.0	370.0	370.0	370.0	370.0	3700.0
46	Sales	(1')	MS						37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	370.0
47																			
48																			
49	STEP 1.5 Profit & Loss table																		
50	Item		Unit			-3	-2	-1	1	2	3	4	5	6	7	8	9	10	Total
51	Sales	(1)	MS						37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0	370.0
52	Operation & Maintenance cost	(5)	MS						7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	70.0
53	Depreciation	(2)	MS						15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	150.0
54	Earning before Interest & Tax (EBIT)	(6)=(1)-(5)-(2)	MS						15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	150.0
55	Interest payment	(32)=(32)+(32')	MS						5.6	4.5	3.3	2.2	1.1	0.0	0.0	0.0	0.0	0.0	16.7
56	Income before Tax	(7)=(6)-(32)	MS						9.4	10.5	11.7	12.8	13.9	15.0	15.0	15.0	15.0	15.0	133.3
57	Tax	(8)=(7)*30%	MS						2.8	3.2	3.5	3.8	4.2	4.5	4.5	4.5	4.5	4.5	40.0
58	Income after tax	(9)=(7)-(8)	MS						6.6	7.4	8.2	8.9	9.7	10.5	10.5	10.5	10.5	10.5	93.3
59																			
60	STEP 2 Calculation of Free Cash Flow (FCF) and Project IRR (PrIRR)																		
61	Item		Unit			-3	-2	-1	1	2	3	4	5	6	7	8	9	10	Total
62	Income after tax	(9)	MS						6.6	7.4	8.2	8.9	9.7	10.5	10.5	10.5	10.5	10.5	93.3
63	Depreciation	(2)	MS						15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	150.0
64	Interest payment	(32)	MS						5.6	4.5	3.3	2.2	1.1	0.0	0.0	0.0	0.0	0.0	16.7
65	Free Cash Flows to Project (FCFP)	(10)=(9)+(2)-(32)	MS						27.2	26.8	26.5	26.2	25.8	25.5	25.5	25.5	25.5	25.5	119.0
66	Project IRR (PrIRR)	9.96%																	
67	WACC=	7.60%																	
68																			
69	STEP 3 Calculation of Equity IRR (EqIRR)																		
70	Item		Unit			-3	-2	-1	1	2	3	4	5	6	7	8	9	10	Total
71	Free Cash Flows to Project (FCFP)	(10)	MS			-40.0	-60.0	-50.0	27.2	26.8	25.5	26.2	25.8	25.5	25.5	25.5	25.5	25.5	119.0
72	Borrowings	(0)	MS				50.0	40.0											90.0
73	Interest payment	(32)	MS						5.6	4.5	3.3	2.2	1.1	0.0	0.0	0.0	0.0	0.0	16.7
74	Principle repayment	(33)=(33)+(33')	MS						18.6	18.6	18.6	18.6	18.6	0.0	0.0	0.0	0.0	0.0	93.0
75	FCF to Equity	(12)=(10)-(0)-(32)-(33)	MS			-40.0	-10.0	-10.0	3.0	3.8	4.6	5.3	6.1	25.5	25.5	25.5	25.5	25.5	90.3
76	Equity IRR (EqIRR)=	11.34%																	
77	Expected Return on Equity (ROE)=	10.00%																	

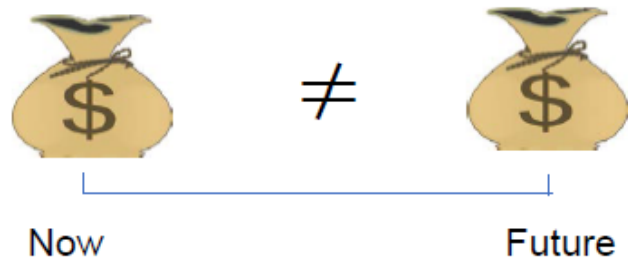
Application of a Financial Model



Present Value of Future Cash Flows

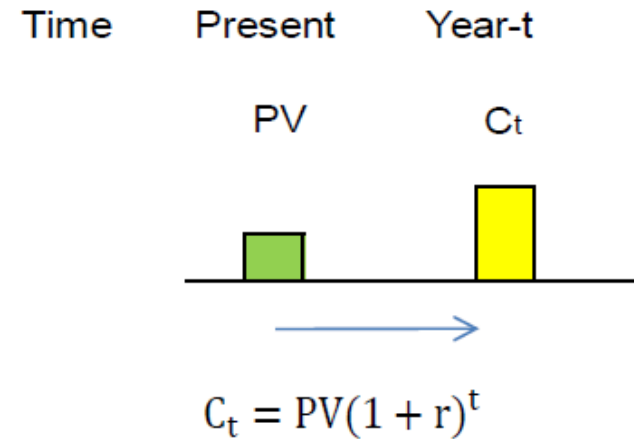
Future value of present cash

US\$ 1 million of today and US\$ 1 million in the future is **not the same value**.



Funds that arise at different times have different values even if the amounts are nominally the same.

We can earn interest if we deposit our money in a bank.



For example, US\$ 1 million of today is worth more than US\$ 1 million of next year. That is because if you put your US\$ 1 million in a bank, it will earn interest and become more than US\$ 1 million next year.

[Example 2.1] Future value of present cash (compound interest calculation)

Example Q1

If the interest rate is 10% a year, how much will be **the current US
\$ 1 million next year?**

Example A1 $1,000,000 \times (1 + 0.1) = 1,000,000 \times 1.1$
 $= 1,100,000$

A. US\$ 1,100,000

[Example 2.1] Future value of present cash (compound interest calculation)

Challenge with pencil and calculator.

Q2 If the interest rate is 10% a year, how much will be **the current US \$ 1 million two (2) years later?**

Q3 If the interest rate is 10% a year, how much will be **the current US \$ 1 million five (5) years later?**

[Example 2.1] Future value of present cash (compound interest calculation)

Answer

Q2 If the interest rate is 10% a year, how much will be the current US \$ 1 million two (2) years later?

$$\begin{aligned}1,000,000 \times (1+0.1) \times (1+0.1) &= 1,000,000 \times (1+0.1)^2 \\ &= 1,000,000 \times 1.21 \\ &= 1,210,000\end{aligned}$$

A. US\$ 1,210,000

Q3 If the interest rate is 10% a year, how much will be the current US \$ 1 million five (5) years later?

$$\begin{aligned}1,000,000 \times (1+0.1) \times (1+0.1) \times (1+0.1) \times (1+0.1) \times (1+0.1) &= 1,000,000 \times (1+0.1)^5 \\ &= 1,000,000 \times 1.61051 \\ &= 1,610,510\end{aligned}$$

A. US\$ 1,610,510

[Example 2.1] Future value of present cash (compound interest calculation)

Challenge with Excel, Sheet Q2.1.

Q1 If the interest rate is 10% a year, how much will be **the current US \$ 1 million next year?**

Q2 If the interest rate is 10% a year, how much will be **the current US \$ 1 million two years later?**

Q3 If the interest rate is 10% a year, how much will be **the current US \$ 1 million five years later?**

How to calculate Compound Interest : $1 \cdot (1+r)^n$			
Year	Compound interest	0.10	
1	$=1 \cdot (1+r)^1$		
2	$=1 \cdot (1+r)^2$		
3	$=1 \cdot (1+r)^3$		
4	$=1 \cdot (1+r)^4$		
5	$=1 \cdot (1+r)^5$		

[Example 2.1] Future value of present cash (compound interest calculation)

Answer

A	B	C	D	E	F	G	H	I
1								
2								
3	How to calculate Compound Interest : $1*(1+r)^n$							
4								
5	Year	Compound interest	0.10					
6	1	$=1*(1+r)^1$	1.10000					
7	2	$=1*(1+r)^2$	1.21000					
8	3	$=1*(1+r)^3$	1.33100					
9	4	$=1*(1+r)^4$	1.46410					
10	5	$=1*(1+r)^5$	1.61051					
11								
12								

$=1*(1+\$C\$5)^{A6}$

$\$C\5 : Cell is fixed

A6 : Cell is floated

Copy Cell (C6) and paste

$=1*(1+\$C\$5)^{A7}$

$=1*(1+\$C\$5)^{A8}$

:

$=1*(1+\$C\$5)^{A10}$

Displayed formula

Year	Compound interest	0.1
1	$=1*(1+r)^1$	$=1*(1+\$C\$5)^{A6}$
2	$=1*(1+r)^2$	$=1*(1+\$C\$5)^{A7}$
3	$=1*(1+r)^3$	$=1*(1+\$C\$5)^{A8}$
4	$=1*(1+r)^4$	$=1*(1+\$C\$5)^{A9}$
5	$=1*(1+r)^5$	$=1*(1+\$C\$5)^{A10}$

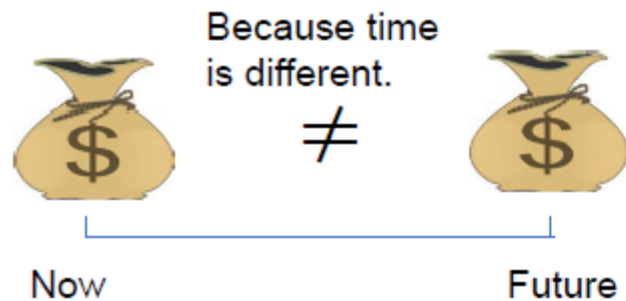
Tips of Excel

Reference	Mark	Function
Relative	C5	Cell address automatically changes by copy & paste.
Absolute	$\$C\5	Cell address is fixed by copy & paste.

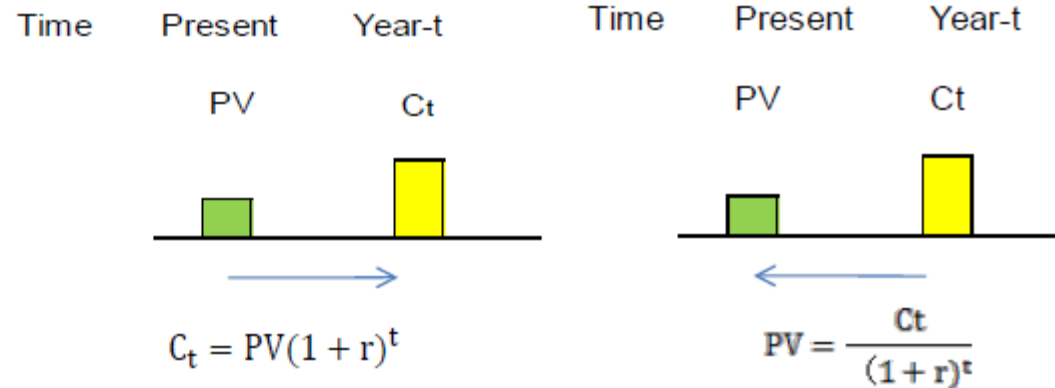
F4 Function button

$\$A\$1 \rightarrow A\$1 \rightarrow \$A1 \rightarrow A1$

Present value of future cash (Discount method)



We should discount a future cash to find the present value of it.



On the contrary, **a future cash has to be discounted** by the rate of interest that would accrue on what it is worth today.

So in finance theory, future funds or future cash flows are converted to what is known as present value. The rate that is used to derive the present value is called the **'discount rate'**.

[Example 2.2] Present value of future cash

Example Q1 What is **the present value of US\$ 1 million which will be received in a year later**? Here the discount rate shall be 10%.

$$\begin{aligned} \text{A1} \quad \text{PV} &= 1,000,000 \div (1 + 0.1) \\ &= \frac{1,000,000}{(1+0.1)} \\ &= \frac{1,000,000}{1.1} \\ &= 909,091 \end{aligned}$$

A. US\$ 909,091

[Example 2.2] Present value of future cash

Challenge with pencil and calculator.

Q2 What is **the present value of US\$ 1 million which will be received in two (2) years later**? Here the discount rate shall be 10%.

Q3 What is **the present value of US\$ 1 million which will be received in five (5) years later**? Here the discount rate shall be 10%.

[Example 2.2] Present value of future cash

Answer

Q2 What is the present value of US\$ 1 million which will be received in two (2) years later? Here the discount rate shall be 10%.

$$\begin{aligned} PV &= 1,000,000 \div (1 + 0.1) \div (1 + 0.1) \\ &= \frac{1,000,000}{(1+0.1)^2} \\ &= \frac{1,000,000}{1.21} \\ &= 826,446 \end{aligned}$$

A. US\$ 826,446

Q3 What is the present value of US\$ 1 million which will be received in five (5) years later? Here the discount rate shall be 10%.

$$\begin{aligned} PV &= 1,000,000 \div (1+0.1) \div (1+0.1) \div (1+0.1) \div (1+0.1) \div (1+0.1) \\ &= \frac{1,000,000}{(1+0.1)^5} \\ &= \frac{1,000,000}{1.61051} \\ &= 620,921 \end{aligned}$$

A. US\$ 620,921

[Example 2.2] Present value of future cash

Challenge with Excel, Sheet Q2.2.

Q1 What is the present value of US\$ 1 million which will be received in a year later?
Here the discount rate shall be 10%.

Q2 What is the present value of US\$ 1 million which will be received in two years later?
Here the discount rate shall be 10%.

Q3 What is the present value of US\$ 1 million which will be received in five years later?
Here the discount rate shall be 10%.

How to calculate discount factor $1/(1+r)^n$			
Year	Discount factor	0.10	
1	$=1/(1+r)^1$		
2	$=1/(1+r)^2$		
3	$=1/(1+r)^3$		
4	$=1/(1+r)^4$		
5	$=1/(1+r)^5$		

[Example 2.2] Present value of future cash

Answer

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5	Year	Discount factor	0.10						
6	1	=1/(1+r)^1	0.90909						
7	2	=1/(1+r)^2	0.82645						
8	3	=1/(1+r)^3	0.75131						
9	4	=1/(1+r)^4	0.68301						
10	5	=1/(1+r)^5	0.62092						
11									
12									

=1/(1+\$C\$5)^A6

\$C\$5 : Cell is fixed

A6 : Cell is floated

Copy Cell (C6) and paste
=1/(1+\$C\$5)^A7
=1/(1+\$C\$5)^A8
:
=1/(1+\$C\$5)^A10

Displayed formula

Year	Discount factor	0.1
1	=1/(1+r)^1	=1/(1+\$C\$5)^A6
2	=1/(1+r)^2	=1/(1+\$C\$5)^A7
3	=1/(1+r)^3	=1/(1+\$C\$5)^A8
4	=1/(1+r)^4	=1/(1+\$C\$5)^A9
5	=1/(1+r)^5	=1/(1+\$C\$5)^A10

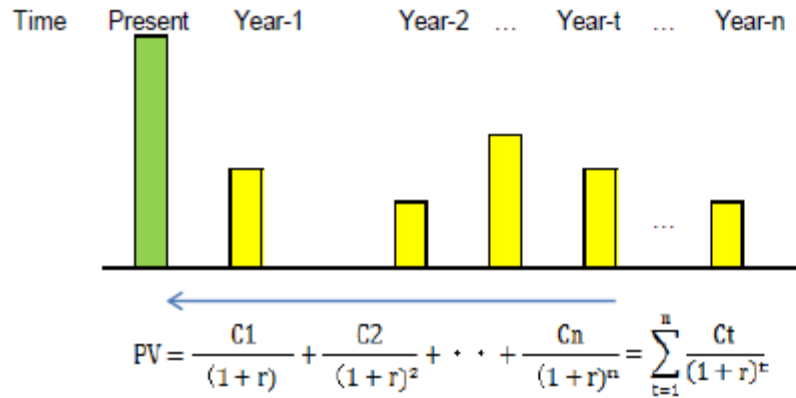
Tips of Excel

Reference	Mark	Function
Relative (Float)	C5	Cell address automatically changes by copy & paste.
Absolute (Fixed)	\$C\$5	Cell address is fixed by copy & paste.

F4 Function button

\$A\$1 → A\$1 → \$A1 → A1

Present value of multiple-year cash flow



In a case where the cash flow for Year-1, Year-2, ...Year-n is expected to be C1, C2...Cn respectively, the aggregate present value of the cash flows is derived with the following formula.

$$PV = \frac{C1}{(1+r)} + \frac{C2}{(1+r)^2} + \dots + \frac{Cn}{(1+r)^n} = \sum_{t=1}^n \frac{Ct}{(1+r)^t}$$

PV: Present value

Ct: Cash flow in Year-t (t = 1, 2, 3 ...)

r: Discount rate

[Example 2.3] Present value of future cash flow

Challenge with pencil and calculator.

Q We can earn US\$ 1,000 every year for 3 years. Find the present value of this cash flow using the table below. The discount rate is assumed to be 10% without change for 3 years.

Year	Year-0	Year-1	Year-2	Year-3	Total present value
Discount rate	10%				
Cash flow		1,000	1,000	1,000	
Discount factor		$1/(1+0.1)$	$1/(1+0.1)^2$	$1/(1+0.1)^3$	
Present value					

[Example 2.3] Present value of future cash flow

Answer

Year	Year-0	Year-1	Year-2	Year-3	Total present value
Discount rate	10%				
Cash flow		1,000	1,000	1,000	
Discount factor		$1/(1+0.1)$	$1/(1+0.1)^2$	$1/(1+0.1)^3$	
		= 0.9091	= 0.8264	= 0.7513	
Present value		909	826	751	2,487

[Example 2.3] Present value of future cash flow

Challenge with Excel, Sheet Q2.3.

Q We can earn US\$ 1,000 every year for 3 years.

Find the present value of this cash flow using the table below.

The discount rate is assumed to be 10% without change for 3 years.

Year	0	1	2	3	Total
Cash flow		1,000	1,000	1,000	
r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	
Discount factor	0.10				
Present Value					

[Example 2.3] Present value of future cash flow

Answer

	A	B	C	D	E	F	G	H	I
1									
2		Year	0	1	2	3	Total		
3		Cash flow		1,000	1,000	1,000			
4			r=	=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3			=SUM(D6:G6)
5		Discount factor	0.10	0.9091	0.8264	0.7513			
6		Present Value		909	826	751	2,487		
7									
8									
9									
10									
11									
12									
13									
14									
15									

Annotations:

- Copy Cell (D5) and paste
 $=1/(1+\$C\$5)^{E2}$
 $=1/(1+\$C\$5)^{F2}$
- Copy cell(D6) and paste
 $=D3*D5$
- $=1/(1+\$C\$5)^{D2}$ (linked to cell C5)
- $=r$:fixed (linked to cell C5)
- $=$ year: floated (linked to cell C5)

Displayed formula

	Year	0	1	2	3	Total
Cash flow			1000	1000	1000	
	r=		=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	
Discount factor	0.1		=1/(1+\$C\$5)^D2	=1/(1+\$C\$5)^E2	=1/(1+\$C\$5)^F2	
Present Value			=D3*D5	=E3*E5	=F3*F5	=SUM(D6:F6)

[Example 2.4] Chicken farm business

Challenge with pencil and calculator.

Q You will receive US\$ 10,000 if you sell your chicken today. But if you keep the chicken, you can sell the eggs and can earn US\$ 3,000 net income every year for four years. Which would be more beneficial to you, selling the chicken today or keeping it for the revenue from eggs? Discuss using the following table. Interest rates are assumed to be constant at 5% during the four years.

(b) Egg sales

(US\$)

Year	Year-0	Year-1	Year-2	Year-3	Year-4	Total present value
Discount rate	$r=5\%$	-	-	-	-	-
Egg sales	-	3,000	3,000	3,000	3,000	-
Discount factor	-					-
Present value	-					

[Example 2.4] Chicken farm business

Answer

(b) Egg sales

(US\$)

Year	Year-0	Year-1	Year-2	Year-3	Year-4	Total present value
Discount rate	$r=5\%$	-	-	-	-	-
Egg sales	-	3,000	3,000	3,000	3,000	-
Discount factor	-	$1/(1+0.05)$ $= 0.9524$	$1/(1+0.05)^2$ $= 0.9070$	$1/(1+0.05)^3$ $= 0.8638$	$1/(1+0.05)^4$ $= 0.8227$	-
Present value	-	2,857	2,721	2,592	2,468	10,638

[Example 2.4] Chicken farm business

Challenge with Excel, Sheet Q2.4.

Year	0	1	2	3	4	Total
Cash flow		3,000	3,000	3,000	3,000	
r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	
Discount factor	0.05					
Present Value						

[Example 2.4] Chicken farm business

Answer

	A	B	C	D	E	F	G	H	I	J
1										
2		Year	0	1	2	3	4	Total		
3		Cash flow		3,000	3,000	3,000	3,000			=SUM(D6:G6)
4			r=	=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4			
5		Discount factor	0.05	0.9524	0.9070	0.8638	0.8227			
6		Present Value		2,857	2,721	2,592	2,468	10,638		
7										
8										
9										
10										
11										
12										
13										
14										
15										

=1/(1+\$C\$5)^D2

= r :fixed

= year: floated

=D3*D5

Copy cell(D6)
and paste

Copy Cell (D5) and paste
=1/(1+\$C\$5)^E2
=1/(1+\$C\$5)^F2
:
=1/(1+\$C\$5)^G2

Displayed formula

Year	0	1	2	3	4	Total
Cash flow		3000	3000	3000	3000	
	r=	=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4	
Discount factor	0.05	=1/(1+\$C\$5)^D2	=1/(1+\$C\$5)^E2	=1/(1+\$C\$5)^F2	=1/(1+\$C\$5)^G2	
Present Value		=D3*D5	=E3*E5	=F3*F5	=G3*G5	=SUM(D6:G6)

[Example 2.5] Present value of business

Challenge with pencil and calculator.

Q Find the present value of the Net-Income after tax for the following business. The tax rate is 25% and the discount rate is constant at 12%.

Year	Year-0	Year-1	Year-2	Year-3	Year-4	Year-5	Total present value
Discount rate	r=12%	-	-	-	-	-	-
Sales	-	2,000	3,000	4,000	5,000	6,000	-
COGS (*)	-	1,000	1,200	1,800	2,000	2,200	-
EBT (**)	-						-
Tax (25%)	-						-
Net Income	-						-
Discount factor	-						-
Present value	-						

* Cost of Goods Sold, ** Earnings before tax

[Example 2.5] Present value of business

Challenge with pencil and calculator.

Q Find the present value of the Net-Income after tax for the following business. The tax rate is 25% and the discount rate is constant at 12%.

Year	Year-0	Year-1	Year-2	Year-3	Year-4	Year-5	Total present value
Discount rate	r=12%	-	-	-	-	-	-
Sales	-	2,000	3,000	4,000	5,000	6,000	-
COGS (*)	-	1,000	1,200	1,800	2,000	2,200	-
EBT (**)	-	1,000					-
Tax (25%)	-	250					-
Net Income	-	750					-
Discount factor	-	$=1/(1+0.12)$					-
Present value	-	669.6					

* Cost of Goods Sold, ** Earnings before tax

[Example 2.5] Present value of business

Answer

Year	Year-0	Year-1	Year-2	Year-3	Year-4	Year-5	Total present value
Discount rate	r=12%	-	-	-	-	-	-
Sales	-	2,000	3,000	4,000	5,000	6,000	-
COGS (*)	-	1,000	1,200	1,800	2,000	2,200	-
EBT (**)	-	1,000	1,800	2,200	3,000	3,800	-
Tax (25%)	-	250	450	550	750	950	-
Net Income	-	750	1,350	1,650	2,250	2,850	-
Discount factor	-	$=1/(1+r)$ $= 0.8929$	$=1/(1+r)^2$ $= 0.7972$	$=1/(1+r)^3$ $= 0.7118$	$=1/(1+r)^4$ $= 0.6355$	$=1/(1+r)^5$ $= 0.5674$	-
Present value	-	670	1,076	1,174	1,430	1,617	5,967

The present value of this business is US\$ 5,967.

[Example 2.5] Present value of business

Challenge with Excel, Sheet Q2.5.

Year	0	1	2	3	4	5	Total
Sales		2,000	3,000	4,000	5,000	6,000	
Cost of sales		1,000	1,200	1,800	2,000	2,200	
EBIT							
Tax	0.25						
Revenue (af. Tax)							
	r=	=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4	=1/(1+r)^5	
Discount factor	0.12						
Present Value							

[Example 2.5] Present value of business

Answer

	A	B	C	D	E	F	G	H	I	J	K
1											
2		=D6-D7	=D8*\$C\$9		=D8-D9						
3											
4											
5		Year	0	1	2	3	4	5	Total		
6		Sales		2,000	3,000	4,000	5,000	6,000			
7		Cost of sales		1,000	1,200	1,800	2,000	2,200			
8		EBIT		1,000	1,800	2,200	3,000	3,800			
9		Tax	0.25	250	450	550	750	950			
10		Revenue (af. Tax)		750	1,350	1,650	2,250	2,850			
11											
12		r=		=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4	=1/(1+r)^5	=SUM(D14:H14)		
13		Discount factor	0.12	0.8929	0.7972	0.7118	0.6355	0.5674			
14		Present Value		670	1,076	1,174	1,430	1,617	5,967		
15											
16											
17											
18				=1/(1+\$C\$13)^D\$5	=D10*D13						
19											
20											

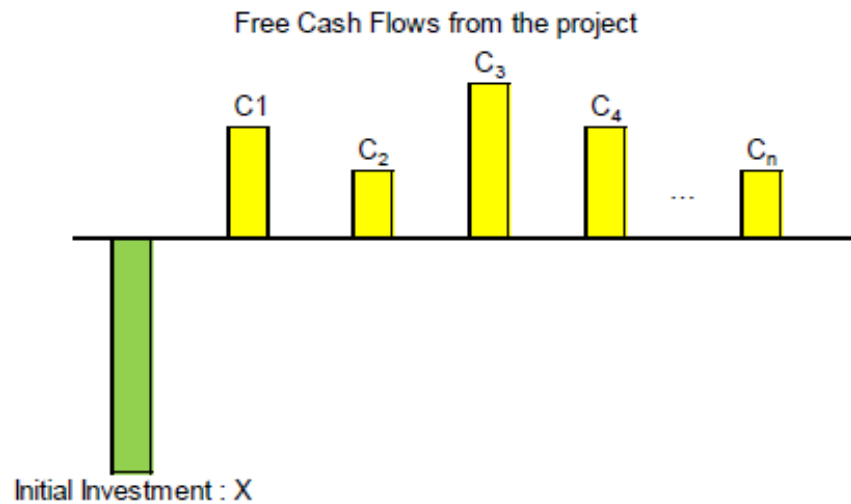
Displayed formula

Year	0	1	2	3	4	5	Total
Sales		2000	3000	4000	5000	6000	
Cost of sales		1000	1200	1800	2000	2200	
EBIT		=D6-D7	=E6-E7	=F6-F7	=G6-G7	=H6-H7	
Tax	0.25	=D8*\$C\$9	=E8*\$C\$9	=F8*\$C\$9	=G8*\$C\$9	=H8*\$C\$9	
Revenue (af. Tax)		=D8-D9	=E8-E9	=F8-F9	=G8-G9	=H8-H9	
r=		=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4	=1/(1+r)^5	
Discount factor	0.12	=1/(1+\$C\$13)^D\$5	=1/(1+\$C\$13)^E\$5	=1/(1+\$C\$13)^F\$5	=1/(1+\$C\$13)^G\$5	=1/(1+\$C\$13)^H\$5	
Present Value		=D10*D13	=E10*E13	=F10*F13	=G10*G13	=H10*H13	=SUM(D14:H14)

Net Present Value (NPV) and Internal Rate of Return (IRR)

Can we start this project?

~ Investment judgement ~



Initial investment : X

Cash flow from the project : $C_1, C_2, \dots, C_n$

Can we start this project?

→ **Investment judgement**

There are several methods of investment judgement.
Most popular ones are:

- **Net Present Value (NPV)** method, and
- **Internal Rate of Return (IRR)** method.
- +
- **Discounted Profitability Index (DPI)** method

[Investment decision]

Net Present Value (NPV) Method

One of the most popular methods to judge an investment is **Net Present Value (NPV) Method**.

Net Present Value (NPV) is calculated by the following formula.

If NPV is positive, then the investment can be justified.

$$\begin{aligned}\text{NPV} &= \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n} - X \\ &= \sum_{t=1}^n \frac{C_t}{(1+r)^t} - X\end{aligned}$$

NPV: Net present value of the investment

C_t: Cash flow in Year-t

X: Investment

r: Discount rate

[Example 3.1] Decision making with Net Present Value (NPV) Method

Challenge with pencil and calculator.

- Q Find the net present value (NPV) of the cash flow of following table using the discount rate of 20%.
Then, judge the investment using NPV.

(US\$ million)

	Year-0	Year-1	Year-2	Year-3	Year-4
Cash flow	-1,200	450	550	700	800
Discount (r =20%)		$1/1.20$	$1/(1.20)^2$	$1/(1.20)^3$	$1/(1.20)^4$
Present value	-1,200				
Net Present Value					

[Example 3.1] Decision making with Net Present Value (NPV) Method

Answer

(US\$ million)

	Year-0	Year-1	Year-2	Year-3	Year-4
Cash flow	-1,200	450	550	700	800
Discount ($r = 20\%$)		$1/1.20$	$1/(1.20)^2$	$1/(1.20)^3$	$1/(1.20)^4$
Present value	-1,200	375.0	381.9	405.1	385.8
Net Present Value	347.8				

The NPV using the discount rate of 20% is US\$ 347.8 million.
Since $NPV \geq 0$, the investment can be justified.

[Example 3.1] Decision making with Net Present Value (NPV) Method

Challenge with Excel, Sheet Q3.1.

Year	0	1	2	3	4	NPV
Cashflow	-1,200	450	550	700	800	
r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	
Discount factor	0.20					
Present Value						

[Example 3.1] Decision making with Net Present Value (NPV) Method

Answer

	A	B	C	D	E	F	G	H	I	J
1										
2										
3										
4										
5		Year	0	1	2	3	4	NPV		
6		Cashflow	-1,200	450	550	700	800			
7		r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$			
8		Discount factor	0.20	0.8333	0.6944	0.5787	0.4823			
9		Present Value	-1,200	375.00	381.94	405.09	385.80	347.84		
10										
11										

Displayed formula

Year	0	1	2	3	4	NPV
Cashflow	-1200	450	550	700	800	
r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	
Discount factor	0.2	$=1/(1+\$C\$8)^D\$5$	$=1/(1+\$C\$8)^E\$5$	$=1/(1+\$C\$8)^F\$5$	$=1/(1+\$C\$8)^G\$5$	
Present Value	$=C6$	$=D6*D8$	$=E6*E8$	$=F6*F8$	$=G6*G8$	$=SUM(C9:G9)$

[Example 3.1] Decision making with Net Present Value (NPV) Method

Change discount rate to 40% instead of 20%, and judge the investment in Excel Sheet Q3.1.

The image displays two Excel spreadsheets side-by-side, illustrating the impact of changing the discount rate on the Net Present Value (NPV) of an investment. Both spreadsheets have columns for Year (0 to 4) and rows for Cashflow, Discount factor, Present Value, and NPV.

Top Spreadsheet (20% Discount Rate):

Year	0	1	2	3	4	NPV
Cashflow	-1,200	450	550	700	800	
Discount factor	0.20	$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	$=SUM(C9:G9)$
Present Value	-1,200	375.00	381.94	405.09	365.80	347.84

Bottom Spreadsheet (40% Discount Rate):

Year	0	1	2	3	4	NPV
Cashflow	-1,200	450	550	700	800	
Discount factor	0.40	$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	$=SUM(C9:G9)$
Present Value	-1,200	321.43	280.61	255.10	208.25	-134.61

Red arrows indicate the change in the discount rate from 20% to 40% and the resulting change in the NPV from 347.84 to -134.61.

When discount rate becomes 40%, then NPV becomes ▲US\$ 134.6 million.

So, the investment loses the justification.

[Investment decision]

Internal Rate of Return (IRR) Method

Another most popular method to judge an investment is **Internal Rate of Return (IRR) Method**.

IRR is a specific discount rate (r) which makes the net present value zero.

IRR means "the yield rate that equalizes the investment and the return".

When the IRR exceeds the hurdle rate, it is judged that investment is justified.

$$NPV = \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n} - X$$

$$X = \sum_{t=1}^n \frac{C_t}{(1+r)^t}$$

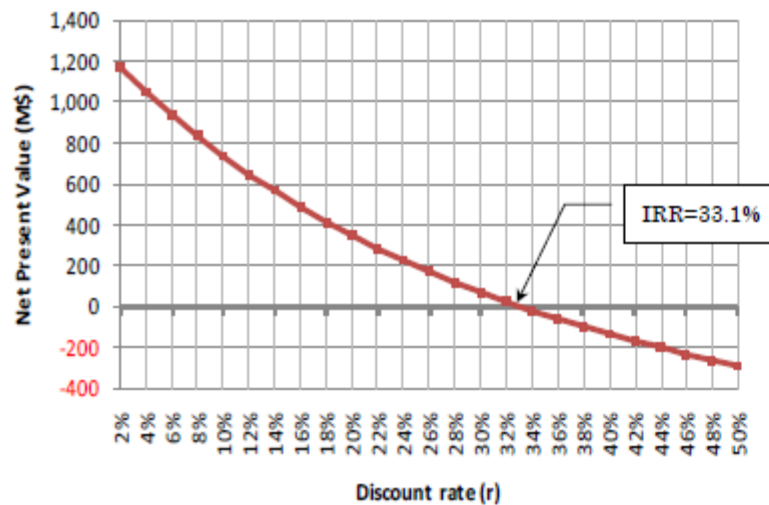
C_t: Cash flow in Year-t

X: Investment

r: Internal rate of return (IRR)

Relation between NPV and IRR

$$NPV = \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n} - X$$



Each project has its own peculiar IRR.

NPV is a function of discount rate (r).

When r is small, NPV is large and positive.

When r becomes larger, NPV becomes smaller.

When r is large enough, NPV is negative.

There is a certain point that NPV = 0.

The r, which makes NPV = 0, is IRR.

Hurdle rate (r_h) = interest rate of borrowing.

If $r_h < IRR$, NPV > 0 → Project is viable.

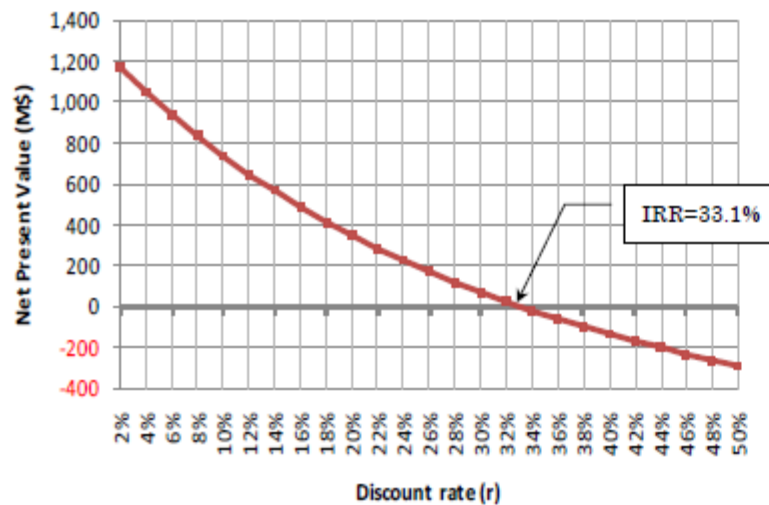
If $r_h > IRR$, NPV < 0 → Project is not viable.

So, we can judge viability of project by calculating **IRR** and comparing IRR with **hurdle rate (interest rate of borrowing)**.

IRR is an index of profitability.

Relation between NPV and IRR

$$NPV = \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n} - X$$



Each project has its own peculiar IRR.

NPV is a function of discount rate (r).

When r is small, NPV is large and positive.

When r becomes larger, NPV becomes smaller.

When r is large enough, NPV is negative.

There is a certain point that NPV = 0.

The r, which makes NPV = 0, is IRR.

Hurdle rate (r_h) = interest rate of borrowing.

If $r_h < IRR$, NPV > 0 → Project is viable.

If $r_h > IRR$, NPV < 0 → Project is not viable.

So, we can judge viability of project by calculating **IRR** and comparing IRR with **hurdle rate (interest rate of borrowing)**.

IRR is an index of profitability.

[Example 3.2] Decision making with IRR Method

Challenge with Excel, Sheet Q3.2.

Calculate IRR for the cashflow of following table by using Excel.

US\$ million

Year	0	1	2	3	4	5	IRR
Cashflow	-100	20	30	40	50	60	

[Example 3.2] Decision making with IRR Method

Answer

To calculate IRR, we use the Excel function [=IRR(range, estimate value)].

US\$ million

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4		Year	0	1	2	3	4	5	IRR		
5		Cashflow	-100	20	30	40	50	60	23.29%		
6											
7											
8											
9											

= IRR(C5:H5)

For this example, IRR = 23.29% is obtained.

[Example 3.3] Comparison of NPV method and IRR method

Challenge with Excel, Sheet Q3.3.

Calculate NPV for the following cashflow by using Excel, and judge the investment. US\$ million

Year	0	1	2	3	4	5	NPV
Cashflow	-150	25	30	40	55	80	
Discount factor	r=	=1/(1+r)	=1/(1+r)^2	=1/(1+r)^3	=1/(1+r)^4	=1/(1+r)^5	
	0.12						
Present Value							

Calculate IRR for the same cashflow by using Excel, and judge the investment. Hurdle rate is 12%.

Year	0	1	2	3	4	5	IRR
Cashflow	-150	25	30	40	55	80	

[Example 3.3] Comparison of NPV method and IRR method

Answer

NPV = US\$ 5.06 million. Since NPV $\geq$ 0, the investment can get green light.

US\$ million							
Year	0	1	2	3	4	5	NPV
Cashflow	-150	25	30	40	55	80	
Discount factor	$r=$ 0.12	$=1/(1+r)$ 0.8929	$=1/(1+r)^2$ 0.7972	$=1/(1+r)^3$ 0.7118	$=1/(1+r)^4$ 0.6355	$=1/(1+r)^5$ 0.5674	
Present Value	-150	22.32	23.92	28.47	34.95	45.39	5.06

IRR = 13.11%. Since NPV $\geq$ 12% (hurdle rate), the investment can get green light.

US\$ million							
Year	0	1	2	3	4	5	IRR
Cashflow	-150	25	30	40	55	80	13.11%

[Example 3.3] Comparison of NPV method and IRR method

Answer

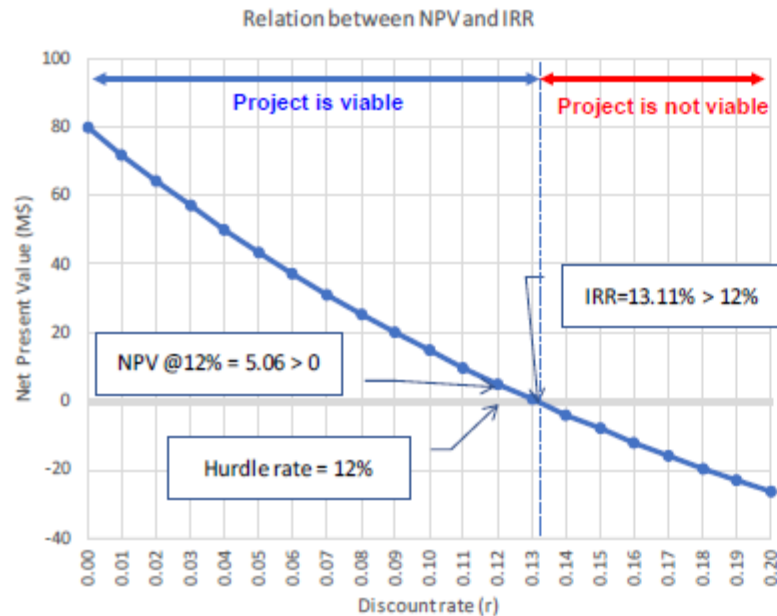
	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5		Year	0	1	2	3	4	5	NPV		
6		Cashflow	-150	25	30	40	55	80			
7		r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	$=1/(1+r)^5$	$=SUM(C9:H9)$		
8		Discount factor	0.12	0.8929	0.7972	0.7118	0.6355	0.5674			
9		Present Value	-150	22.32	23.92	28.47	34.95	45.39	5.06		
10				$=D6*D8$							
11											
12		Year	0	1	2	3	4	5	IRR		
13		Cashflow	-150	25	30	40	55	80	13.11%		
14									$=IRR(C13:H13)$		
15											
16											

Displayed formula

Year	0	1	2	3	4	5	NPV
Cashflow	-150	25	30	40	55	80	
r=		$=1/(1+r)$	$=1/(1+r)^2$	$=1/(1+r)^3$	$=1/(1+r)^4$	$=1/(1+r)^5$	
Discount factor	0.12	$=1/(1+\$C\$8)*D\$5$	$=1/(1+\$C\$8)*E\$5$	$=1/(1+\$C\$8)*F\$5$	$=1/(1+\$C\$8)*G\$5$	$=1/(1+\$C\$8)*H\$5$	
Present Value	=C6	=D6*D8	=E6*E8	=F6*F8	=G6*G8	=H6*H8	=SUM(C9:H9)
Year	0	1	2	3	4	5	IRR
Cashflow	-150	25	30	40	55	80	=IRR(C13:H13)

Relation between NPV and IRR in Example 3.3

$$NPV = \frac{C1}{(1+r)} + \frac{C2}{(1+r)^2} + \dots + \frac{Cn}{(1+r)^n} - X$$



NPV at 12% discount rate = US\$ 5.06 million.
NPV > 0 → Project is viable.

IRR = 13.11% > 12% of hurdle rate. → Project is viable

[Exercise 3.4] Quiz of IRR

Quiz

Q1 Can IRR exceed 100%?

Q2 Can IRR be zero?

Q3 Can IRR be negative?

[Exercise 3.4] Quiz of IRR

Answer

Q1 Yes.

Q2 Yes.

Q3 Yes.

Year	0	1	2	3	4	5	IRR
Case-1	-100	110	110	110	110	110	107.11%
Case-2	-100	20	20	20	20	20	0.00%
Case-3	-100	15	15	15	15	15	-8.88%

Free Cash Flow

Income Statement

Example of Income Statement

Item	M\$	Remarks
Sales	100	a
COGS (Cost of Goods Sold)	60	$b=c+d+e+f$
Raw material expense	30	c
Depreciation expense	10	d
Wages	15	e
O&M cost	5	f
EBIT (Earnings Before Interest & Tax)	40	$g=a-b$
Interest Expense	5	h
EBT (Earnings Before Tax)	35	$i=g-h$
Tax ($t=40\%$)	14	$j=i*40\%$
Net Income	21	$k=i-j$

[Example 4.1] Practice to make an Income Statement

There is a project that needs **initial investment of US\$ 500,000**.

From this project, **you can expect sales** of US\$ 200,000 for the first year, US\$ 250,000 for the second year, US\$ 300,000 for the third year, US\$ 350,000 for the fourth year, and US\$ 400,000 for the fifth year.

The expected expenses for this project are;

- (i) **Raw material expenses** are US\$ 20,000 for the first year, US\$ 25,000 for the second year, US\$ 30,000 for the third year, US\$ 35,000 for the fourth year, and US\$ 40,000 for the fifth year.
- (ii) **Wages** are fixed at US\$ 50,000 each year,
- (iii) **Operation and maintenance cost** is constant at US\$ 10,000 every year.
- (iv) No need to consider depreciation expenses (set to zero).
- (v) No need to consider payment of interest or repayment of loans. (set to zero).
- (vi) **Tax rate is 30%**.

('000US\$)

	Year-0	Year-1	Year-2	Year-3	Year-4	Year-5
Investment	-500					
Sales		200	250	300	350	400
Raw material		20	25	30	35	40
Wages		50	50	50	50	50
O&M		10	10	10	10	10



Please make an Income Statement.

[Example 4.1] Creation of Simple Income Statement

Q1 Complete Income Statement with pencil and calculator

Item	Year	0	1	2	3	4	5
Sales	a		200	250	300	350	400
COGS (Cost of Goods Sold)	b=c+d+e+f						
Raw material expense	c						
Depreciation expense	d						
Wages	e						
O&M cost	f						
EBIT (Earnings Before Interest & Tax)	g=a-b						
Interest Expense	h						
EBT (Earnings Before Tax)	i=g-h						
Tax (t=30%)	j=i*30%						
Net Income	k=i-j						

[Example 4.1] Creation of Simple Income Statement

Q2 Calculate NPV of the net income cash flow with pencil and calculator.
Q3 Calculate IRR of the net income cash flow by Excel.
Q4 Judge the investment.

Answers of Net Income from previous page.

Item	Year	0	1	2	3	4	5
Net Income	$k=i-j$						
Present value of Net Income	$r = 10\%$	-500					
Net Present Value =							
Cash flow of Net Income		-500					
IRR =							

[Example 4.1] Creation of Simple Income Statement

Answer

Item	Year	0	1	2	3	4	5
Sales	a		200	250	300	350	400
COGS (Cost of Goods Sold)	b=c+d+e+f		80	85	90	95	100
Raw material expenses	c		20	25	30	35	40
Depreciation	d		0	0	0	0	0
Wages	e		50	50	50	50	50
O&M costs	f		10	10	10	10	10
EBIT (Earnings Before Interest & Tax)	g=a-b		120	165	210	255	300
Interest Expense	h		0	0	0	0	0
EBT (Earnings Before Tax)	i=g-h		120	165	210	255	300
Tax (t=30%)	j=i*30%		36	49.5	63	76.5	90
Net Income	k=i-j		84	115.5	147	178.5	210
Present Value of Net Income	r = 10%	-500	76.4	95.5	110.4	121.9	130.4
Net Present Value =	34.6						
Cash flow of Net Income for IRR		-500	84	115.5	147	178.5	210
IRR =	12.3%						

[Example 4.1] Creation of Simple Income Statement

Challenge it again with Excel, Sheet Q4.1.

Item	Year	0	1	2	3	4	5	
Sales	a		200	250	300	350	400	
COGS (Cost of Goods Sold)	b=c+d+e+f							
Raw material expenses	c							
Depreciation	d							
Wages	e							
O&M costs	f							
EBIT (Earnings Before Interest & Tax)	g=a-b							
Interest Expense	h							
EBT (Earnings Before Tax)	i=g-h							
Tax (t=30%)	j=i*30%							
Net Income	k=i-j							
								IRR
Cash flow		-500						
	discount rate (r) =	10.0%						NPV
Present value		-500						

[Example 4.1] Creation of Simple Income Statement

Answer

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3		Item	Year	0	1	2	3	4	5		
4		Sales	a		200	250	300	350	400		
5		COGS (Cost of Goods Sold)	b=c+d+e+f		80	85	90	95	100		
6		Raw material expenses	c		20	25	30	35	40		
7		Depreciation	d	=Sum(E6:E9)	0	0	0	0	0		
8		Wages	e	=E4-E5	50	50	50	50	50		
9		O&M costs	f		10	10	10	10	10		
10		EBIT (Earnings Before Interest & Tax)	g=a-b		120	165	210	255	300		
11		Interest Expense	h	=E10-E11	0	0	0	0	0		
12		EBT (Earnings Before Tax)	i=g-h		120	165	210	255	300		
13		Tax (t=30%)	j=i*30%	=E12*0.3	36	49.5	63	76.5	90		
14		Net Income	k=i-j	=E12-E13	84	115.5	147	178.5	210		
15											
16		Cash flow	=E14	-500	84	115.5	147	178.5	210		
17											
18			discount rate (r) =	10.0%							
19		Present value		-500	76.4	95.5	110.4	121.9	130.4		
20											
21											
22											
23											
24											
25											
26											

Copy and paste

=IRR(D16:I16)

IRR

12.3%

NPV

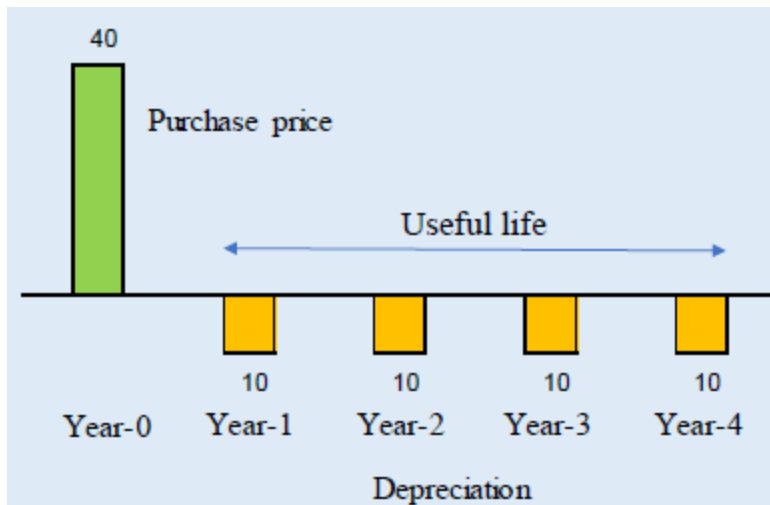
=E16/(1+\$D\$18)^E3

=SUM(D19:I19)

\$D\$18 : Discount rate cell is fixed

E3 : Year cell is floated

Depreciation expense

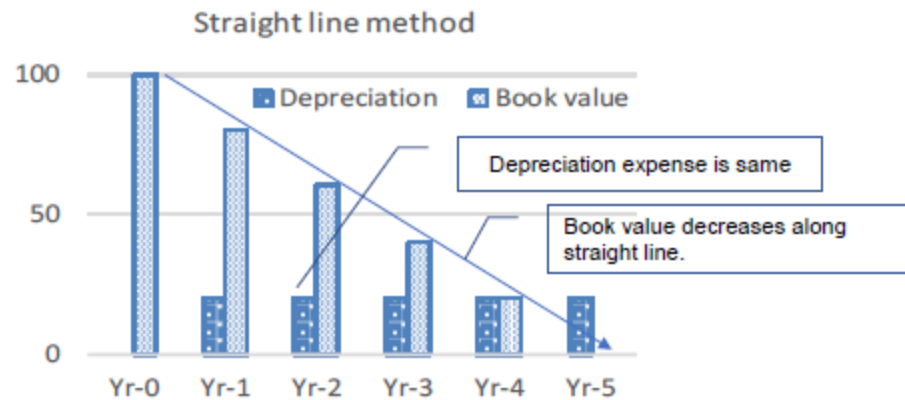


Depreciation is an accounting method to allocate the cost of a fixed asset over its useful life.

For example, when we purchase a machine, which can be used for four (4) years, at US\$ 40,000, we depreciate it at US\$ 10,000 every year over the four years.

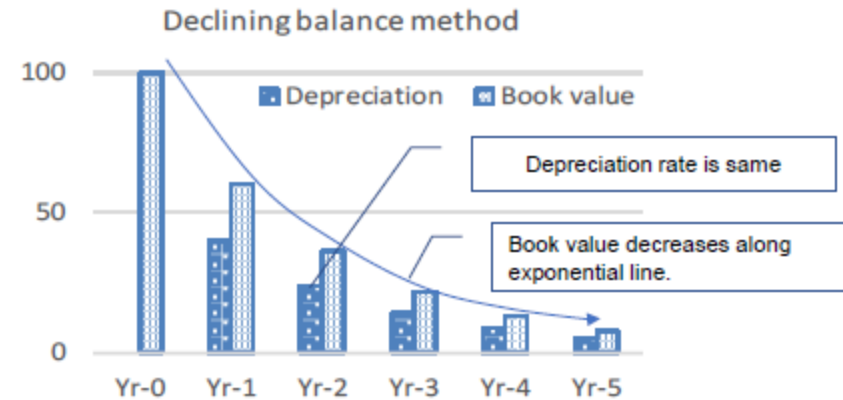
Straight line method and Declining balance method

There are two common depreciation methods; one is "straight line method" and another is "declining balance method".



$$\text{Depreciation expense} = (\text{Acquisition cost} - \text{Residual value}) / \text{Useful life}$$

If residual value is zero, then
 $\text{Depreciation per annum} = \text{Acquisition cost} / \text{Useful life}$



$$\text{Depreciation expense} = (\text{Acquisition cost} - \text{Accumulated depreciation}) \times \text{Depreciation rate}$$

[Example 4.2] Depreciation expense

Calculate depreciation expense with pencil and calculator.

We purchased **US\$ 500,000** machinery as an initial investment.
The **useful life** of this machinery is **5 years**.

Calculate the annual **depreciation expense** and the **book value** at the end of each year using the **straight-line method**.

(US\$ 1,000)

Year	0	1	2	3	4	5
Book Value at the end of year	500					
Depreciation expense	-					

[Example 4.2] Depreciation expense

Answer

(US\$ 1,000)

Year	0	1	2	3	4	5
Book Value at the end of year	500	400	300	200	100	0
Depreciation expense	-	100	100	100	100	100

[Example 4.2] Depreciation expense

Challenge it again with Excel, Sheet Q4.2.

We purchased **US\$ 500,000 machinery** as an initial investment.
The **useful life** of this machinery is **5 years**.

Calculate the annual **depreciation expense** and the **book value** at the end of each year using the **straight-line method**.

(US\$ 1,000)						
Item	0	1	2	3	4	5
Book Value (at year-end)	500					
Straight line method						
Depreciation	-					

[Example 4.2] Depreciation expense

Answer

	A	B	C	D	E	F	G	H	I
1									
2			=C4-D6						
3		Item	0	1	2	3	4	5	
4		Book Value	500	400	300	200	100	0	
5		Straight line method							
6		Depreciation	-	100	100	100	100	100	
7			=C\$4/5						
8									
9									
10									
11									
12									
13									

Annotations:

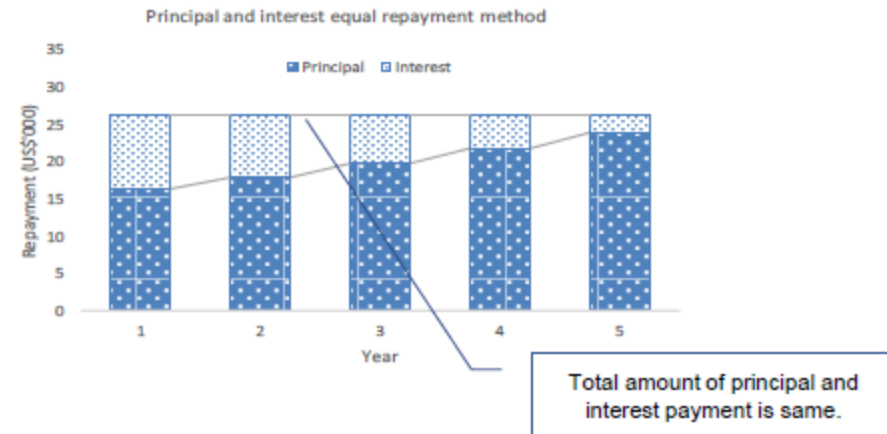
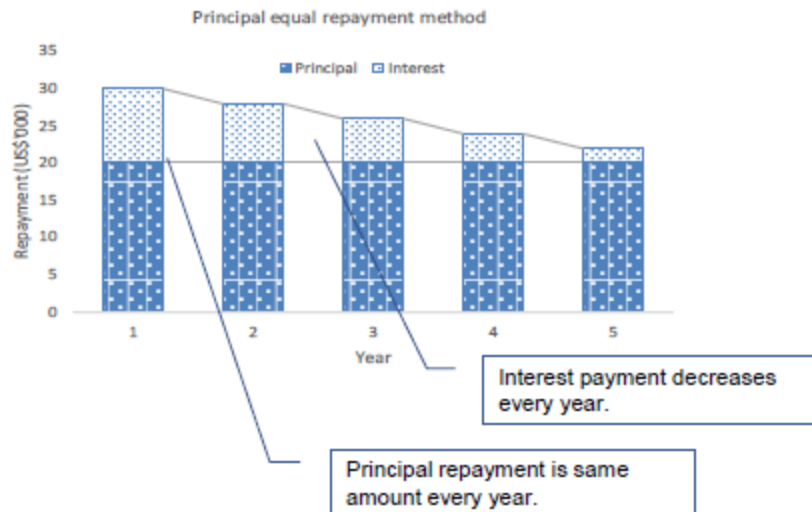
- Box 1: =C4-D6 (points to cell C2)
- Box 2: =C\$4/5 (points to cell C7)
- Box 3: Copy and paste (points to cell F7)
- Box 4: \$C\$4 : Initial investment is fixed (points to cell C4)
- Box 5: 5 : Life of machine (points to cell C7)

Displayed formula

Item	0	1	2	3	4	5
Book Value (at year-end)	500	=C4-D6	=D4-E6	=E4-F6	=F4-G6	=G4-H6
Straight line method						
Depreciation	-	=C\$4/5	=C\$4/5	=C\$4/5	=C\$4/5	=C\$4/5

Principal equal repayment method and Principal & interest equal repayment method

There are two common repayment methods; one is "Principal equal repayment method" and another is "Principal and interest equal repayment method".



To calculate repayment amount in "principal and interest equal repayment method", we need to use Excel function:

Principal repayment

= PPMT(interest rate, repayment year, repayment period, borrowing amount)

Interest payment

= IPMT(interest rate, repayment year, repayment period, borrowing amount)

[Example 4.3] Repayment of Borrowings

Create a repayment schedule with pencil and calculator.

We borrowed **US\$ 250,000** from a bank for a project.
The interest rate is **10% per annum** and the **repayment period is 5 years**.
Create a repayment schedule using the principal equal repayment method.

Year	0	1	2	3	4	5
Borrowing	250	-	-	-	-	-
Repayment period	5					
Interest rate (r)	10%					
Principal repayment	-					
Interest payment	-					
Total payment	-					
Borrowing balance (at year-end)	250					

[Example 4.3] Repayment of Borrowings

Answer

Year	0	1	2	3	4	5
Borrowing	250	-	-	-	-	-
Repayment period	5	Same amount (equal principle repayment method)				
Interest rate (r)	10%					
Principal repayment	Calculation order - 1	-	50	50	50	50
Interest payment	Calculation order - 3	-	25	20	15	10
Total payment	Calculation order - 4	-	75	70	65	60
Borrowing balance (at year-end)	Calculation order - 2	250	200	150	100	50

Interest is charged to the borrowing balance of previous year-end.

[Example 4.3] Repayment of Borrowings

Challenge it again with Excel, Sheet Q4.3.

We borrowed US\$ 250,000 from a bank for a project.
The interest rate is 10% per annum and the repayment period is 5 years.
Create a repayment schedule using the equal principal repayment method.

Item	0	1	2	3	4	5
Borrowing	250					
Repayment period	5					
Interest rate (r)	10%					
Principal repayment	-					
Interest payment	-					
Total repayment	-					
Borrowing balance (at year-end)	250					

Simple method is not effective when repayment period changes from 5 years to 7 years, etc.

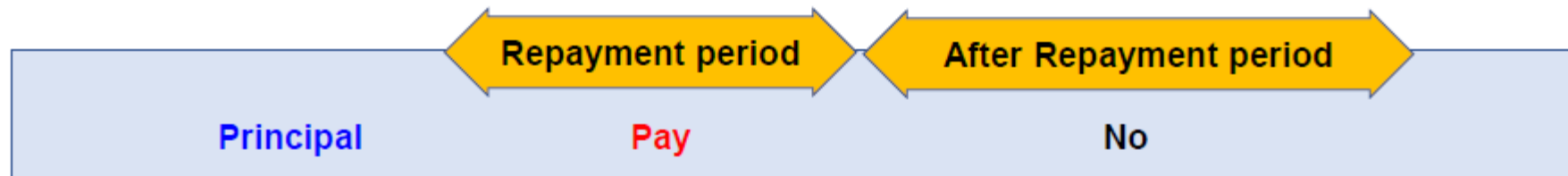
We need some wisdom.....

If the **current year is within the repayment period** (5 years)

→ **We need calculation.**

If the **current year is out of the repayment period** (5 years)

→ **All answers should be zero (0).**



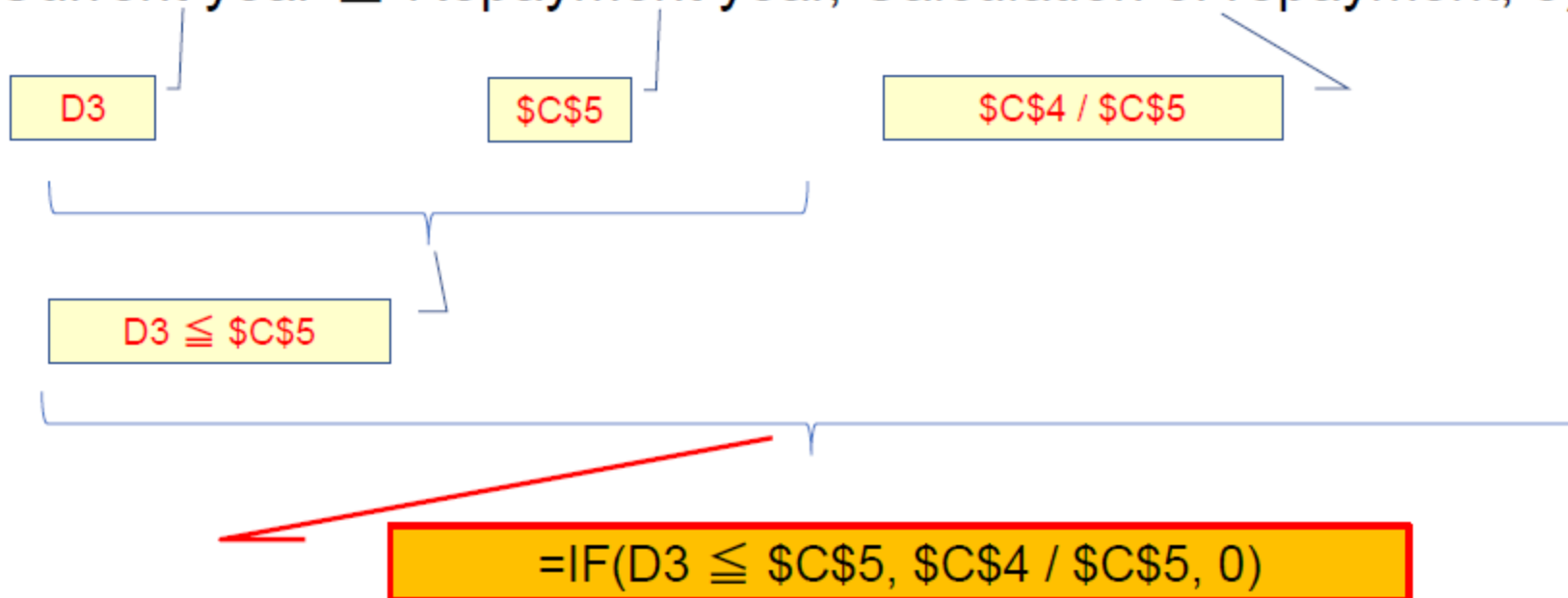
= IF (Condition, Yes ans., No ans.)

= IF (current year <= repayment period, Calculation, 0)

Wisdom ~ use IF function ~

= IF(Condition, Yes case answer, No case answer)

= IF(Current year $\leq$ Repayment year, Calculation of repayment, 0)



[Example 4.3] Repayment of Borrowings

Answer						
Item	0	1	2	3	4	5
Borrowing	250					
Repayment period	5					
Interest rate (r)	10%					
Principal repayment	-	50	50	50	50	50
Interest payment	-	25	20	15	10	5
Total repayment	-	75	70	65	60	55
Borrowing balance (at year-end)	250	200	150	100	50	0

Item	0	1	2	3	4	5
Borrowing	250					
Repayment period	5					
Interest rate (r)	0.1					
Principal repayment	-	=IF(D3<=\$C\$5,\$C\$4/\$C\$5,0)	=IF(E3<=\$C\$5,\$C\$4/\$C\$5,0)	=IF(F3<=\$C\$5,\$C\$4/\$C\$5,0)	=IF(G3<=\$C\$5,\$C\$4/\$C\$5,0)	=IF(H3<=\$C\$5,\$C\$4/\$C\$5,0)
Interest payment	-	=IF(D3<=\$C\$5,C10*\$C\$6,0)	=IF(E3<=\$C\$5,D10*\$C\$6,0)	=IF(F3<=\$C\$5,E10*\$C\$6,0)	=IF(G3<=\$C\$5,F10*\$C\$6,0)	=IF(H3<=\$C\$5,G10*\$C\$6,0)
Total repayment	-	=D7+D8	=E7+E8	=F7+F8	=G7+G8	=H7+H8
Borrowing balance (at year-end)	250	=C10-D7	=D10-E7	=E10-F7	=F10-G7	=G10-H7

=IF(D3<=\$C\$5, ...,0)
During repayment period → calculation, otherwise → 0)

\$C\$4/\$C\$5:
Borrowing/Repayment period

Copy and paste

C10*\$C\$6:
Balance * Int. rate

=IF(D3<=\$C\$5, ...,0)
During repayment period → calculation, otherwise → 0)

Displayed formula

Use 'IF Function' so that we can change
repayment period (for ex. 5 years → 4 years).

[Example 4.3] Repayment of Borrowings

Challenge it again with Excel, Sheet Q4.3.

From the view point of lenders, calculate an IRR using the result of repayment schedule.

Item	0	1	2	3	4	5
Borrowing	250					
Repayment period	5					
Interest rate (r)	10%					
Principal repayment	-					
Interest payment	-					
Total repayment	-					
Borrowing balance (at year-end)	250					

(Check)	0	1	2	3	4	5	IRR
Lender's IRR	-250						

Calculate Lender's IRR :
Initial investment is payment of principal.
Annual income is interest and repayment of principal.

Next, change the borrowing schedule and calculate Lender's IRR in each case.

Case	1	2	3	4
Repayment period (years)	3	4	5	2
Interest rate (r)	10%	12%	8%	5%
Lender's IRR				

[Example 4.3] Repayment of Borrowings

Answer

	A	B	C	D	E	F	G	H	I	J
1										
2										
3		Item	0	1	2	3	4	5		
4		Borrowing	250							
5		Repayment period	5							
6		Interest rate (r)	10%							
7		Principal repayment	-	50	50	50	50	50		
8		Interest payment		25	20	15	10	5		
9		Total repayment		75	70	65	60	55		
10		Borrowing balance (at year-end)	250	200	150	100	50	0		
11										
12		(for check)	0	1	2	3	4	5	IRR	
13		Lender's IRR	-250	75	70	65	60	55	10.0%	
14										
15										
16										

Lender's IRR = 10% (= interest rate)

= IRR(C13:H13)

Next, change the borrowing schedule and calculate Lender's IRR in each case.

Case	1	2	3	4
Repayment period (years)	3	4	5	2
Interest rate (r)	10%	12%	8%	5%
Lender's IRR	10%	12%	8%	5%

[Tips]

Lender's IRR is always equal to the interest rate of borrowings.

So, when you make a repayment schedule, create Lender's IRR and check it whether it is equal to the interest rate. You can know your repayment schedule is correctly created or not.

[Example 4.4(1)] Calculate IDC

Challenge with a pencil and a calculator.

Interest rate = 10%

	Year	Yr-(-4)	Yr-(-3)	Yr-(-2)	Yr-(-1)	As of COD
	Borrowing each year (a)	100	200	300	400	-
	Cumulative borrowing (w/o IDC) (a')	100	300	600	1,000	1,000
Borrowing in Yr-(-4)	Borrowing balance (b)=(a)x(1+r)^n	100				
Borrowing in Yr-(-3)	Borrowing balance (b)=(a)x(1+r)^n		200			
Borrowing in Yr-(-2)	Borrowing balance (b)=(a)x(1+r)^n			300		
Borrowing in Yr-(-1)	Borrowing balance (b)=(a)x(1+r)^n				400	
Total	Total borrowing balance (with IDC) (c)					
	IDC (cum.) (d)=(c)-(a')					
	IDC of each year (e)					-

[Example 4.4(1)] Calculate IDC

Answer

Interest rate = 10%

	Year	Yr-(-4)	Yr-(-3)	Yr-(-2)	Yr-(-1)	As of COD
	Borrowing each year (a)	100	200	300	400	-
	Cumulative borrowing (w/o IDC) (a')	100	300	600	1,000	1,000
Borrowing in Yr-(-4)	Borrowing balance (b)=(a)x(1+r)^n	100	110 (=100*1.1)	121 (=100*1.1 ²)	133.1 (=100*1.1 ³)	133.1
Borrowing in Yr-(-3)	Borrowing balance (b)=(a)x(1+r)^n		200	220 (=200*1.1)	242 (=200*1.1 ²)	242
Borrowing in Yr-(-2)	Borrowing balance (b)=(a)x(1+r)^n			300	330 (=300*1.1)	330
Borrowing in Yr-(-1)	Borrowing balance (b)=(a)x(1+r)^n				400	400
Total	Total borrowing balance (with IDC) (c)	100	310	641	1,105.1	1,105.1
	IDC (cum.) (d)=(c)-(a')	—	10	41	105.1	105.1
	IDC of each year (e)	—	10	31	64.1	—

[Example 4.4(1)] Calculate IDC

Challenge with Excel “Q4.4 sheet”

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2			Multi-year borrowings									
3												
4			Borrowing	1000	M\$							
5			-4 year Borrowing	100	M\$							
6			-3 year Borrowing	200	M\$							
7			-2 year Borrowing	300	M\$							
8			-1 year Borrowing	400	M\$							
9			Loan period	5	years							
10			Interest rate	10%	a.n.							
11			Principal equal repayment									
12												
13			Borrowing Amount & IDC Calculation									
14			Year		Unit		-4	-3	-2	-1	Total	
15			Borrowing Requirement (annual)	(a)	M\$		100	200	300	400		
16			Borrowing Requirement (cum.)		M\$		100	300	600	1,000		
17												
18			Borrowing in Year (-4)	(b)=(left cell)x(1+r)	M\$		100.0					
19			Borrowing in Year (-3)	(b)=(left cell)x(1+r)	M\$			200.0				
20			Borrowing in Year (-2)	(b)=(left cell)x(1+r)	M\$				300.0			
21			Borrowing in Year (-1)	(b)=(left cell)x(1+r)	M\$					400.0	400.0	
22												
23			Borrowing with IDC Total	(c)=Σ (b)	M\$		100.0	200.0	300.0	400.0	400.0	
24			IDC (cum.)	(d)=(c)-(a)	M\$			-100.0	-300.0	-600.0	-	
25			IDC annually accrued		M\$			-100.0	-200.0	-300.0	-600.0	
26												

Data

Input formula in all cells.

Input formula

[Example 4.4(1)] Calculate IDC

Answer

[Tips] Technic for IDC calculation

= Left cell * (1+ Interest rate)

= Left cell * (1+\$D\$10)

and Copy & Paste it to right cells

11	Principal equal repayment								
12									
13	Borrowing Amount & IDC Calculation								
14	Year	Unit	-4	-3	-2	-1	Total		
15	Borrowing Requirement (annual)	(a) MS	100	200	300	400	1,000		
16	Borrowing Requirement (cum.)	MS	100	300	600	1,000			
17									
18	Borrowing in Year (-4)	(b)=(left cell)x(1+r) MS	100.0	110.0	121.0	133.1	133.1		
19	Borrowing in Year (-3)	(b)=(left cell)x(1+r) MS		200.0	220.0	242.0	242.0		
20	Borrowing in Year (-2)	(b)=(left cell)x(1+r) MS			300.0	330.0	330.0		
21	Borrowing in Year (-1)	(b)=(left cell)x(1+r) MS				400.0	400.0		
22									
23	Borrowing with IDC Total	(c)= Σ (b) MS	100.0	310.0	641.0	1,105.1	1,105.1		
24	IDC (cum.)	(d)=(c)-(a) MS		10.0	41.0	105.1	-		
25	IDC annually accrued	MS		10.0	31.0	64.1	105.1		
26									

= Left cell *(1+\$D\$10)

Calculation of Repayment Schedule

Equal principal repayment method		Commercial Operation Date (COD)					
		0	1	2	3	4	5
Principal repayment (a)	$128.2 = 641 / 5$		128.2	128.2	128.2	128.2	128.2
Interest payment (b)	<Interest rate = 10%>		$64.1 \times 10\%$	$51.3 \times 10\%$	38.5	25.6	12.8
Total repayment (b)+(c)			192.3	179.5	166.7	153.8	141.0
Outstanding at year end (d)-(a)		641	512.8	384.6	256.4	128.2	0
Lender's Cash Flow (b)+(c)		-641	192.3	179.5	166.7	153.8	141.0
Lender's IRR		10.00%					

Lender's IRR = Interest rate

[Example 4.4(2)] Calculation of Repayment Schedule

Challenge with Excel “Q4.4 sheet”

[illegible]

[Example 4.4(2)] Calculation of Repayment Schedule

Answer

11	Principal equal repayment																	
12																		
13	Borrowing Amount & IDC Calculation																	
14	Year		Unit	-4	-3	-2	-1	Total										
15	Borrowing Requirement (annual)	(a)	M\$	100	200	300	400	1,000										
16	Borrowing Requirement (cum.)		M\$	100	300	600	1,000											
17																		
18	Borrowing in Year (-4)	(b)=(left cell)x(1+r)	M\$	100.0	110.0	121.0	133.1	133.1										
19	Borrowing in Year (-3)	(b)=(left cell)x(1+r)	M\$		200.0	220.0	242.0	242.0										
20	Borrowing in Year (-2)	(b)=(left cell)x(1+r)	M\$			300.0	330.0	330.0										
21	Borrowing in Year (-1)	(b)=(left cell)x(1+r)	M\$				400.0	400.0										
22																		
23	Borrowing with IDC Total	(c)=Σ (b)	M\$	100.0	310.0	641.0	1,105.1	1,105.1										
24	IDC (cum.)	(d)=(c)-(a)	M\$		10.0	41.0	105.1	-										
25	IDC annually accrued		M\$		10.0	31.0	84.1	105.1										
26																		
27																		
28	Borrowing Repayment Schedule																	
29	Year			-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	Total
30	Principal repayment		M\$					221.0	221.0	221.0	221.0	221.0	0.0	0.0	0.0	0.0	0.0	1105.1
31	Interest payment		M\$					110.5	88.4	66.3	44.2	22.1	0.0	0.0	0.0	0.0	0.0	331.5
32	Total repayment		M\$					331.5	309.4	287.3	265.2	243.1	0.0	0.0	0.0	0.0	0.0	1438.6
33	Balance of the year end		M\$				1105.1	884.1	683.1	442.0	221.0	0.0	0.0	0.0	0.0	0.0	0.0	-
34																		
35	Lender's Cash Flow		M\$	-100.0	-200.0	-300.0	-400.0	331.5	309.4	287.3	265.2	243.1	0.0	0.0	0.0	0.0	0.0	
36	Lender's IRR	10.00%	OK															

= Left cell *(1+\$D\$10)

Use 'IF Function' so that we can change repayment period (for ex. 5 years → 4 years)

\$D\$9: Repayment period

\$J\$33: Total borrowing amount

=IF(K29<=\$D\$9,\$J\$33/\$D\$9,0)

=IF(K29<=\$D\$9,J33*\$D\$10,0)

J33: Balance of previous year end

Free Cash Flow

To calculate IRR, we have used the word of “cash flow” in the previous sections for simplicity reasons. But it should be “**Free cash flow (FCF)**” instead of “cash flow” in strict meaning. Free cash flow is also called “**Free cash flow to the project (FCFP)**” when we need to distinguish this free cash flow from the “**Free cash flow to equity (FCFE)**” which will be explained in the next section.

In this workbook, the FCFP is defined as the following formula.

$$\begin{aligned} &\text{Free cash flow to the project (FCFP)} \\ &= \text{Net Income (after tax)} + \text{Depreciation expense} + \text{Interest repayment} \\ &\quad - \text{Investment} \end{aligned}$$

[Example 4.5]

Free cash flow to the project (FCFP)

Calculate FCFP with pencil and calculator.

Calculate Free Cash Flow to the Project (FCFP) from this Income Statement (US\$ 1,000)

Item	Year-1	Year-2
Sales	100	120
COGS (Cost of Goods Sold)	60	65
Raw material expenses	30	35
Depreciation	10	10
Wages	15	15
O&M costs	5	5
EBIT (Earnings Before Interest & Tax)	40	55
Interest Expense	5	4
EBT (Earnings Before Tax)	35	51
Tax (t=40%)	14	20.4
Net Income	21	29.6
Free Cash Flow to the Project (FCFP)		

[Example 4.5]

Free cash flow to the project (FCFP)

Answer

Using formula of

FCFP = Net Income (after tax) + Depreciation cost + Interest payment – Investment,
and Investment = 0,

Free cash flow to the project in Year-1 is US\$ 36,000, and
Free cash flow to the project in Year-2 is US\$ 43,600.

(US\$ 1,000)

Item	Year-1	Year-2
Net Income	21	29.6
Depreciation expense	10	10
Interest Expense	5	4
Free Cash Flow to the Project (FCFP)	36	43.6

Item	Year-1	Year-2
Sales	100	120
COGS (Cost of Goods Sold)	60	65
Raw material expenses	30	35
Depreciation	10	10
Wages	15	15
O&M costs	5	5
EBIT (Earnings Before Interest & Tax)	40	55
Interest Expense	5	4
EBT (Earnings Before Tax)	35	51
Tax (t=40%)	14	20.4
Net Income	21	29.6
Free Cash Flow to the Project (FCFP)		

[Example 4.6] Free Cash Flow to Project (FCFP) and Project IRR

[Conditions]

There is a project that needs **initial investment of US\$ 1,000,000**.

From this project, you can **expect sales of US\$ 500,000** every year for 5 years.

The expected expenses for this project are;

- (i) **Raw material** expenses are **US\$ 20,000 (constant)**,
- (ii) **Wages** are **US\$ 50,000 (constant)**, and
- (iii) **Operation and maintenance costs** are **US\$ 40,000 (constant)** in every year.
- (iv) Initial investment of machinery is **US\$ 1,000,000**. The life year of the machinery is 5 years.
Depreciation is straight line method.

All of initial investment is funded by the borrowings from a commercial bank.

The interest rate is 8% per annum and the repayment period is 5 years.

The repayment method is **principal equal repayment**.

Tax rate is 30%.

[Example 4.6] Free Cash Flow to Project (FCFP) and Project IRR

Challenge it again with Excel, Sheet Q4.6.

	A	B	C	D	E	F	G	H	I	J
1										
2		P&L calculation section							(1,000 US\$)	
3		Item	Year	0	1	2	3	4	5	
4		Sales	a		500	500	500	500	500	
5		COGS (Cost of Goods Sold)	b=c+d+e+f							
6		Raw material expenses	c							
7		Depreciation	d=m							
8		Wages	e							
9		O&M costs	f							
10		EBIT (Earnings Before Interest & Tax)	g=a-b							
11		Interest Expense	h=t							
12		EBT (Earnings Before Tax)	i=g-h							
13		Tax (t=30%)	j=i*30%							
14		Net Income	k=i-j							
15										
16		Depreciation calculation section								
17		Item	Year	0	1	2	3	4	5	
18		Book Value	m	1000						
19		Straight line method								
20		Depreciation	n	-						
21										
22		Repayment calculation section								
23		Item	0	1	2	3	4	5		
24		Borrowing	o	1000						
25		Repayment period	(p)	5						
26		Interest rate (r)	(r)	8%						
27		Principal repayment	s=o/(p)	-						
28		Interest payment	t=v*(r)	-						
29		Total repayment	u=s+t	-						
30		Borrowing balance (at year-end)	v	1000						
31										
32		(Check)		0	1	2	3	4	5	IRR
33		Lender's IRR	w=u	-1000						
34										
35		Cashflow calculation section								
36		Item	Year	0	1	2	3	4	5	IRR
37		Free Cash flow to the Project (FCFP)	x=k+d+h	-1000						
38										

Q1 Calculate the depreciation cost of every year between Row-18 and Row-20.

Q2 Create a repayment schedule of borrowings between Row-24 and Row-30.
(Calculate Lender's IRR in Row-33 and compare it with interest rate.)

Q3 Create a five-year income statement for this project and calculate free cash flow of each year.

Q4 Calculate the internal rate of return (IRR) of this project.

[Example 4.6] Free Cash Flow to Project (FCFP) and Project IRR

Answer

Displayed formula

	A	B	C	D	E	F	G	H	I	J	K
1											
2		P&L calculation section							(1,000 US\$)		
3		Item	Year	0	1	2	3	4	5		
4		Sales	a		500	500	500	600	500		
5		COGS (Cost of Goods Sold)	b=c+d+e+f		310	310	310	310	310		
6		Raw material expenses	c		20	20	20	20	20		
7		Depreciation	d=m		200	200	200	200	200		
8		Wages	e		50	50	50	50	50		
9		O&M costs	f		40	40	40	40	40		
10		EBIT (Earnings Before Interest & Tax)	g=a-b		190.0	190.0	190.0	190.0	190.0		
11		Interest Expense	h=t		80.0	64.0	48.0	32.0	16.0		
12		EBT (Earnings Before Tax)	i=g-h		110.0	126.0	142.0	158.0	174.0		
13		Tax (t=30%)	j=i*30%		33.0	37.8	42.6	47.4	52.2		
14		Net Income	k=i-j		77.0	88.2	99.4	110.6	121.8		
15											
16		Depreciation calculation section									
17		Item	Year	0	1	2	3	4	5		
18		Book Value	m		1000	800	600	400	200		
19		Straight line method									
20		Depreciation	n		-	200	200	200	200		
21											
22		Repayment calculation section									
23		Item	Year	0	1	2	3	4	5		
24		Borrowing	o		1000						
25		Repayment period	(p)		5						
26		Interest rate (r)	(r)		8%						
27		Principal repayment	s=o/(p)		-	200.0	200.0	200.0	200.0		
28		Interest payment	t=v*(r)		-	80.0	64.0	48.0	32.0		
29		Total repayment	u=s+t		-	280.0	264.0	248.0	232.0		
30		Borrowing balance (at year-end)	v		1000	800.0	600.0	400.0	200.0		
31											
32		(Check)			0	1	2	3	4	5	IRR
33		Lender's IRR	w=u		-1000	290.0	264.0	248.0	232.0	216.0	8.0%
34											OK
35		Cashflow calculation section									
36		Item	Year	0	1	2	3	4	5	IRR	
37		Free Cash flow to the Project (FCFP)	x=k+d+h		-1000	357.0	352.2	347.4	342.6	337.8	22.0%
38											

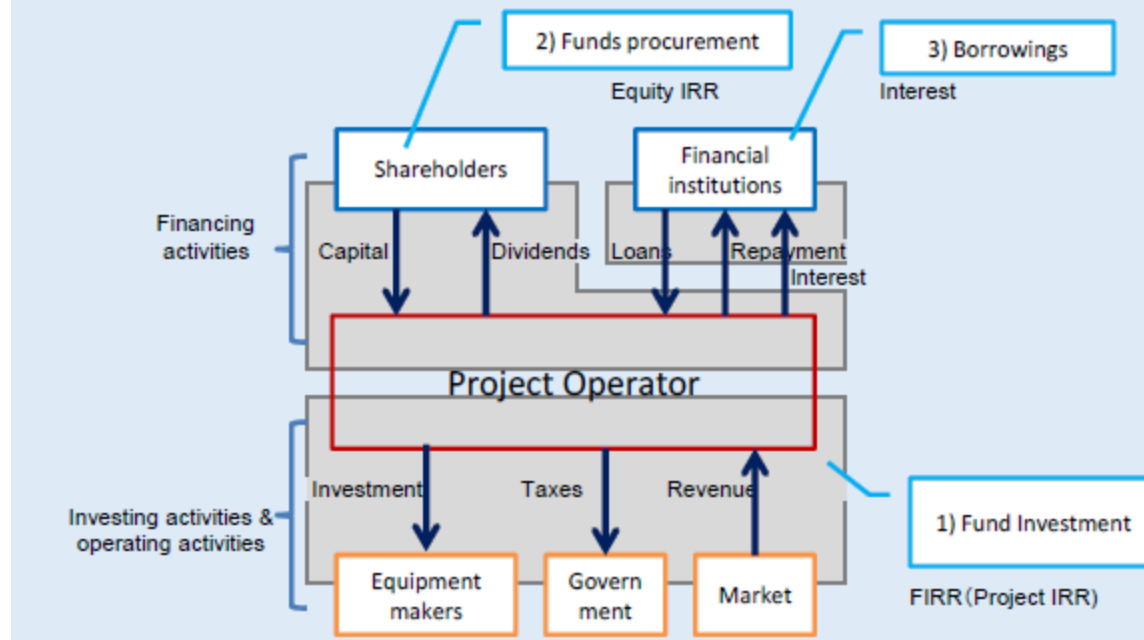
	A	B	C	D	E	F
1						
2		P&L calculation section				
3		Item	Year	0	1	2
4		Sales	a		510	=E4
5		COGS (Cost of Goods Sold)	b=c+d+e+f		=SUM(E5:E9)	=SUM(F5:F9)
6		Raw material expenses	c		21	20
7		Depreciation	d=m		=E20	=F20
8		Wages	e		51	50
9		O&M costs	f		41	40
10		EBIT (Earnings Before Interest & Tax)	g=a-b		=E4-E5	=F4-F5
11		Interest Expense	h=t		=E21	=F21
12		EBT (Earnings Before Tax)	i=g-h		=E10-E11	=F10-F11
13		Tax (t=30%)	j=i*30%		=E12*0.3	=F12*0.3
14		Net Income	k=i-j		=E12-E13	=F12-F13
15						
16		Depreciation calculation section				
17		Item	Year	0	1	2
18		Book Value	m		=D18-E20	=E18-F20
19		Straight line method				
20		Depreciation	n		=D18/5	=E18/5
21						
22		Repayment calculation section				
23		Item	Year	0	1	2
24		Borrowing	o		1000	
25		Repayment period	(p)		5	
26		Interest rate (r)	(r)		0.08	
27		Principal repayment	s=o/(p)		=F(E23<=>\$D\$25:\$D\$24/5D\$25,1)	=F(F23<=>\$D\$25:\$D\$24/5D\$25,1)
28		Interest payment	t=v*(r)		=F(E23<=>\$D\$25:D30*\$D\$26,1)	=F(F23<=>\$D\$25:E30*\$D\$26,1)
29		Total repayment	u=s+t		=E27+E28	=F27+F28
30		Borrowing balance (at year-end)	v		=D24	=E30-E27
31						
32		(Check)			0	1
33		Lender's IRR	w=u		=D24*(1-1)	=E29
34						
35		Cashflow calculation section				
36		Item	Year	0	1	2
37		Free Cash flow to the Project (FCFP)	x=k+d+h		=D18*(1-1)	=E14+E7+E11
38						

Project IRR is 22.0%.

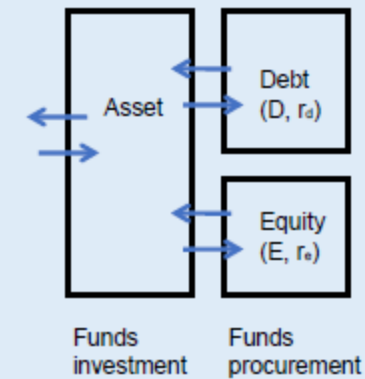
Three IRRs surrounding project

There are three (3) cash flows surrounding the project.

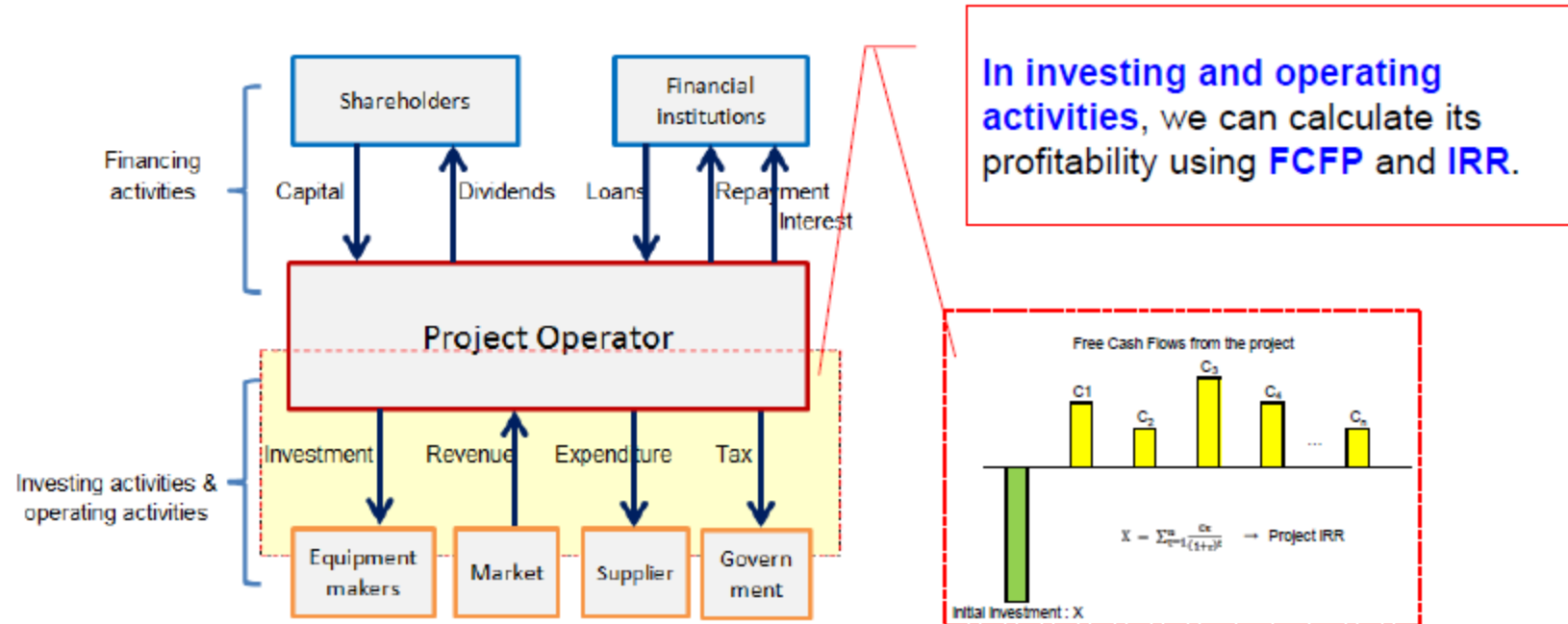
- One is cash flow from and to the project operator,
- Another is cash flow from and to the banks, and
- The other is cash flow from and to equity suppliers.



This is same structure as balance sheet (BS).

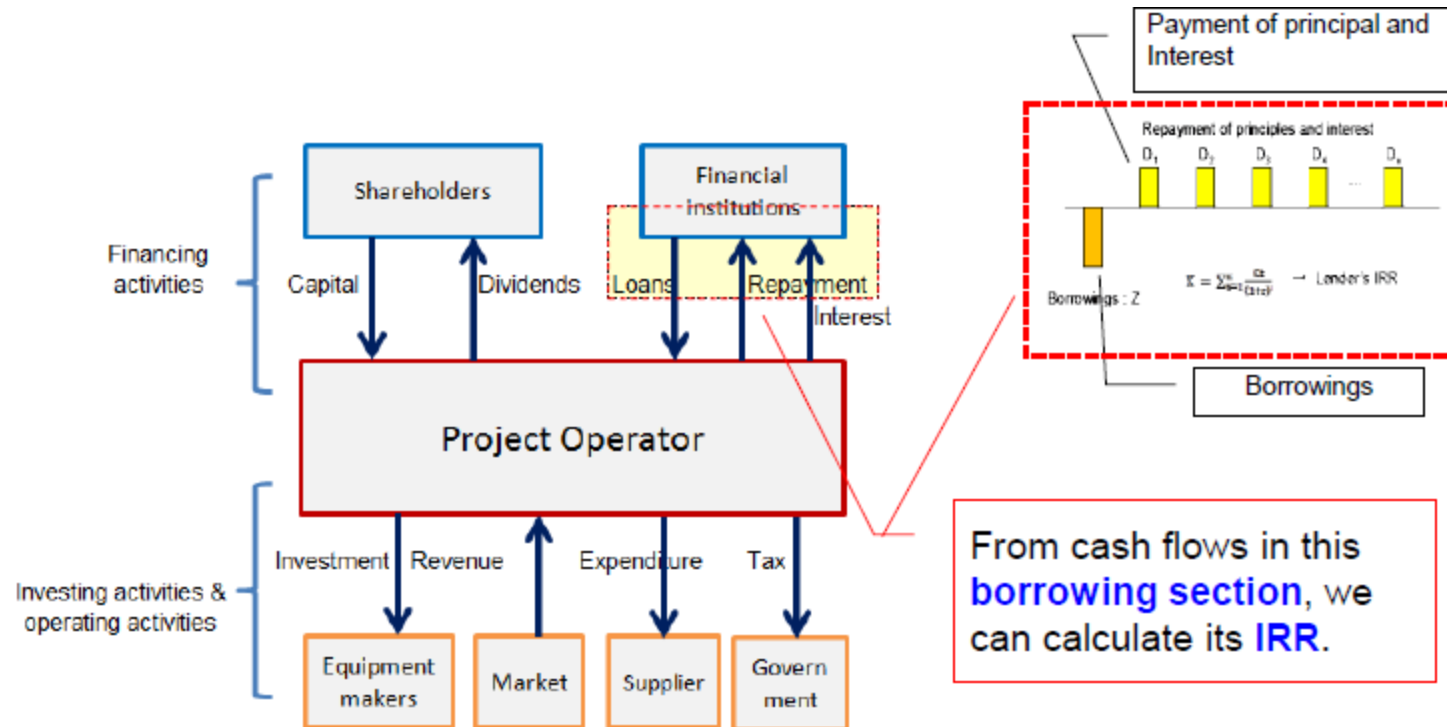


Project IRR (Financial IRR; FIRR)



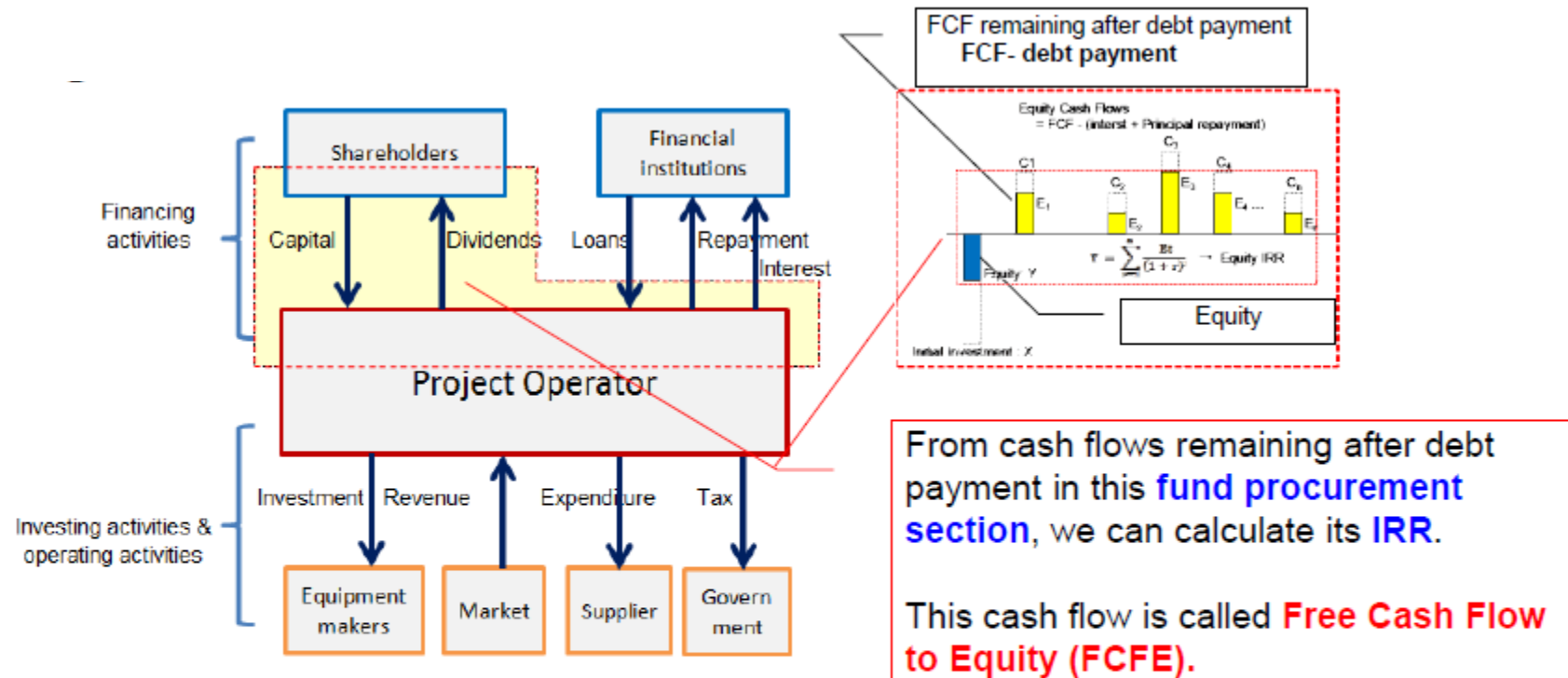
The **IRR** calculated in this section is called **Project IRR (PrIRR)** or **Financial IRR (FIRR)**.

Lender's IRR



The **IRR** calculated in this borrowing section is called **Lenders IRR**, and always **equals to interest rate**.

Equity IRR



Free Cash Flow to Equity (FCFE) = FCFP – Principal repayment – Interest payment

The **IRR** calculated in this shareholders section is called **Equity IRR**.

[Example 4.7] Free Cash Flow to Equity (FCFE) and Equity IRR

Calculate FCFE and Equity IRR with Excel, Sheet Q4.7.

	A	B	C	D	E	F	G	H	I	J	K
1											
2		P&L calculation section							(1,000 US\$)		
3		Item	Year	0	1	2	3	4	5		
4		Sales	a		500	500	500	500	500		
5		COGS (Cost of Goods Sold)	b=c+d+e+f		310	310	310	310	310		
6		Raw material expenses	c		20	20	20	20	20		
7		Depreciation	d=m		200	200	200	200	200		
8		Wages	e		50	50	50	50	50		
9		OSM costs	f		40	40	40	40	40		
10		EBIT (Earnings Before Interest & Tax)	g=a-b		190.0	190.0	190.0	190.0	190.0		
11		Interest Expense	h=i		20.0	16.0	12.0	8.0	4.0		
12		EBT (Earnings Before Tax)	i=g-h		170.0	174.0	178.0	182.0	186.0		
13		Tax (t=30%)	j=i*30%		51.0	52.2	53.4	54.6	55.8		
14		Net Income	k=i-j		119.0	121.8	124.6	127.4	130.2		
15											
16		Depreciation calculation section									
17		Item	Year	0	1	2	3	4	5		
18		Book Value	m	1000	800	600	400	200	0		
19		Straight line method									
20		Depreciation	n	-	200	200	200	200	200		
21											
22		Repayment calculation section									
23		Item	Year	0	1	2	3	4	5		
24		Borrowing	o	250							
25		Repayment period	(p)	5							
26		Interest rate (r)	(r)	8%							
27		Principal repayment	s=o/(p)	-	50.0	50.0	50.0	50.0	50.0		
28		Interest payment	t=v*(r)	-	20.0	16.0	12.0	8.0	4.0		
29		Total repayment	u=s+t	-	70.0	66.0	62.0	58.0	54.0		
30		Borrowing balance (at year-end)	v	250	200.0	150.0	100.0	50.0	0.0		
31											
32		(Check)		0	1	2	3	4	5	IRR	
33		Lender's IRR	w=u	-250	70.0	66.0	62.0	58.0	54.0	8.0%	OK
34											
35		Cashflow calculation section									
36		Item	Year	0	1	2	3	4	5	PvIRR	
37		Free Cash flow to the Project (FCFP)	x=k+d+h	-1000	339.0	337.8	336.6	335.4	334.2	20.4%	
38											
39		Item	Year	0	1	2	3	4	5	EqIRR	
40		Free Cash flow to Equity (FCFE)	y=x-s-t	-750							
41											
42											
43											
44											

In the case of Example 4.7, the share holder provided US\$ 750,000, and a bank provided US\$ 250,000 to the project.

Q1 Calculate cash flow to the **Equity (FCFE)** of every year in Row-40.

Q2 Calculate **Equity IRR** of this project.

Input formula to calculate Equity IRR

[Example 4.7] Free Cash Flow to Equity (FCFE) and Equity IRR

Answer

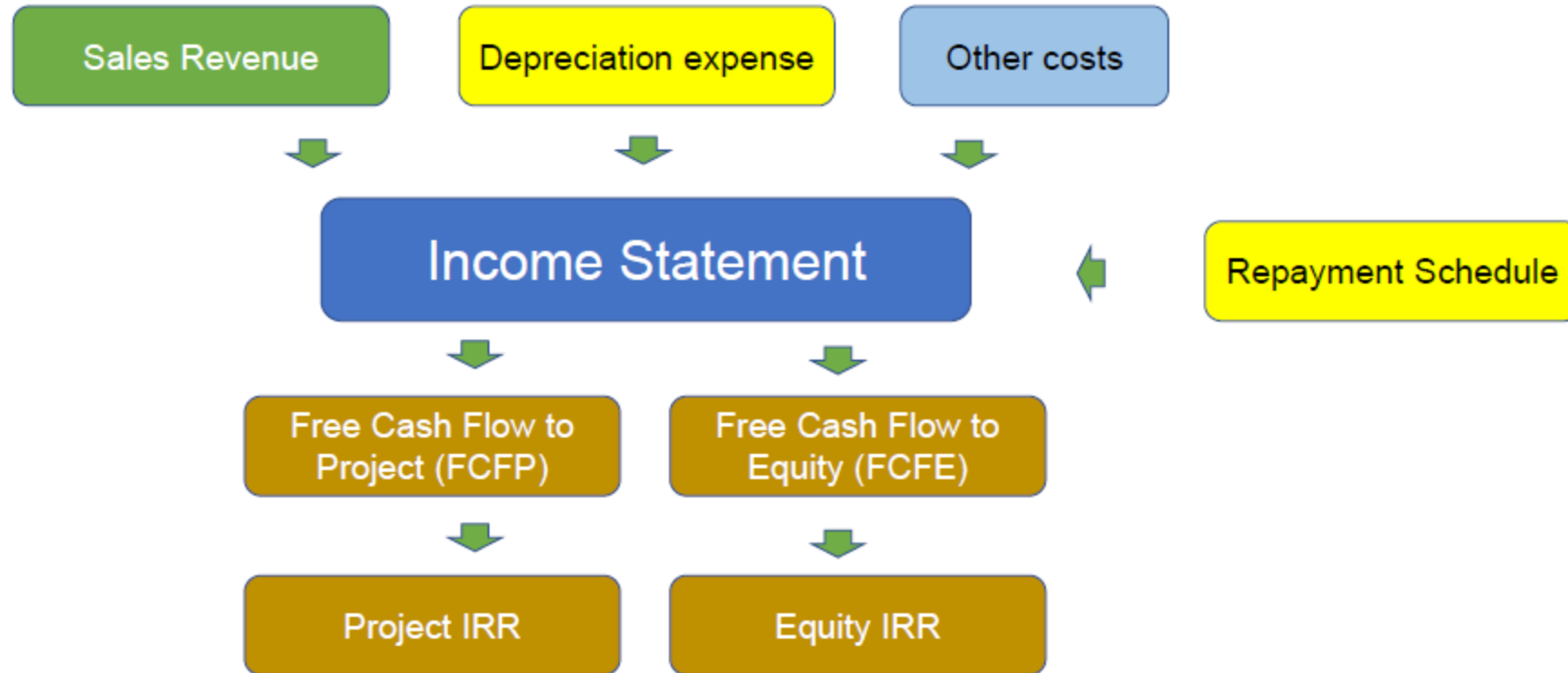
	A	B	C	D	E	F	G	H	I	J	K
1											
2		P&L calculation section							(1,000 US\$)		
3		Item	Year	0	1	2	3	4	5		
4		Sales	a		500	500	500	500	500		
5		COGS (Cost of Goods Sold)	b=c+d+e+f		310	310	310	310	310		
6		Raw material expenses	c		20	20	20	20	20		
7		Depreciation	d=m		200	200	200	200	200		
8		Wages	e		50	50	50	50	50		
9		O&M costs	f		40	40	40	40	40		
10		EBIT (Earnings Before Interest & Tax)	g=a-b		190.0	190.0	190.0	190.0	190.0		
11		Interest Expense	h=t		20.0	16.0	12.0	8.0	4.0		
12		EBT (Earnings Before Tax)	i=g-h		170.0	174.0	178.0	182.0	186.0		
13		Tax (t=30%)	j=i*30%		51.0	52.2	53.4	54.6	55.8		
14		Net Income	k=i-j		119.0	121.8	124.6	127.4	130.2		
15		Depreciation calculation section									
16		Item	Year	0	1	2	3	4	5		
17		Book Value	m		1000	800	600	400	200		
18		Straight line method									
19		Depreciation	n		-	200	200	200	200	200	
20											
21		Repayment calculation section									
22		Item	Year	0	1	2	3	4	5		
23		Borrowing	o		250						
24		Repayment period	(p)		5						
25		Interest rate (r)	(r)		8%						
26		Principal repayment	s=o/(p)			50.0	50.0	50.0	50.0	50.0	
27		Interest payment	t=r*(r)			20.0	16.0	12.0	8.0	4.0	
28		Total repayment	u=s+t			70.0	66.0	62.0	58.0	54.0	
29		Borrowing balance (at year-end)	v		250	200.0	150.0	100.0	50.0	0.0	
30											
31		(Check)		0	1	2	3	4	5	IRR	
32		Lender's IRR	w=u		-250	70.0	66.0	62.0	58.0	54.0	8.0%
33											OK
34		Cashflow calculation section									
35		Item	Year	0	1	2	3	4	5	PIRR	
36		Free Cash flow to the Project (FCFP)	x=k+d+h		-1000	339.0	337.8	336.6	335.4	334.2	20.4%
37											
38		Item	Year	0	1	2	3	4	5	EqIRR	
39		Free Cash flow to Equity (FCFE)	y=x-s-u		-750.0	269.0	271.8	274.6	277.4	280.2	24.0%
40											
41											

Displayed formula

	A	B	C	D	E
1					
2		P&L calculation section			
3		Item	Year	0	1
4		Sales	a		500
5		COGS (Cost of Goods Sold)	b=c+d+e+f		=SUM(E6:E9)
6		Raw material expenses	c		20
7		Depreciation	d=m		=E20
8		Wages	e		50
9		O&M costs	f		40
10		EBIT (Earnings Before Interest & Tax)	g=a-b		=E4-E5
11		Interest Expense	h=t		=E28
12		EBT (Earnings Before Tax)	i=g-h		=E10-E11
13		Tax (t=30%)	j=i*30%		=E12*0.3
14		Net Income	k=i-j		=E12-E13
15		Depreciation calculation section			
16		Item	Year	0	1
17		Book Value	m		1000
18		Straight line method			
19		Depreciation	n		=D18-E20
20					
21		Repayment calculation section			
22		Item	Year	0	1
23		Borrowing	o		250
24		Repayment period	(p)		5
25		Interest rate (r)	(r)		0.08
26		Principal repayment	s=o/(p)		=IF(E23<=\$D\$25,\$D\$24/\$D\$25,0)
27		Interest payment	t=r*(r)		=IF(E23<=\$D\$25,D30*\$D\$26,0)
28		Total repayment	u=s+t		=E27+E28
29		Borrowing balance (at year-end)	v		=D24
30					
31		(Check)		0	1
32		Lender's IRR	w=u		=D24*(-1)
33					
34		Cashflow calculation section			
35		Item	Year	0	1
36		Free Cash flow to the Project (FCFP)	x=k+d+h		=E14+E7+E11
37					
38		Item	Year	0	1
39		Free Cash flow to Equity (FCFE)	y=x-s-u		=D37-D29+D24
40					
41					

Equity IRR is 24.0%.

Bird-eye View of creating a Financial Model



Feasibility Judgment

Project IRR $\geq$ Hurdle rate of operator, or WACC

Hurdle rate: Some company have their own hurdle rate for decision (ex.12%).

WACC: Weighted Average Cost of Capital

$$WACC = \frac{E}{(E + D)} r_e + \frac{D}{(E + D)} r_d$$

E: Equity or funds from shareholders

r_e : Cost of equity funding (expected rate of return)

D: Debt or borrowings from financial institutions

r_d : Cost of debt funding (interest rate)

Balance sheet

Asset	Debt (D, r_d)
	Equity (E, r_e)
Funds investment	Funds procurement

Lender's IRR = Interest rate of borrowing ← no need to judge

Equity IRR $\geq$ Expected rate of return on equity (r_e)

r_e : depends on equity provider's management philosophy

Exercise: WACC Calculation

WACC: Weighted Average Cost of Capital

$$WACC = \frac{E}{(E + D)} r_e + \frac{D}{(E + D)} r_d$$

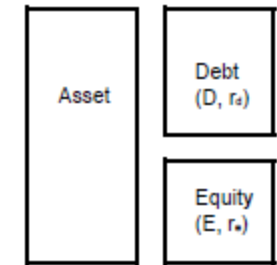
E: Equity or funds from shareholders

r_e : Cost of equity funding (expected rate of return)

D: Debt or borrowings from financial institutions

r_d : Cost of debt funding (interest rate)

Balance sheet



Funds
investment

Funds
procurement

Total Project Cost	100	M\$	(E+D)
Loan amount	70	M\$	(D)
Equity amount	30	M\$	(E)
Interest rate	6%		(r _d)
Cost of Equity	12%		(r _e)
Loan portion			(D)/(E+D)
Equity portion			(E)/(E+D)
WACC			Loan portion * r _d + Equity portion * r _e

Exercise: WACC Calculation

WACC: Weighted Average Cost of Capital

$$WACC = \frac{E}{(E + D)} r_e + \frac{D}{(E + D)} r_d$$

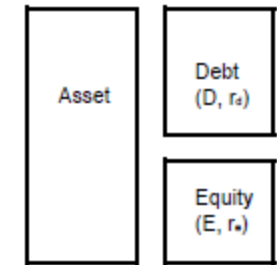
E: Equity or funds from shareholders

r_e : Cost of equity funding (expected rate of return)

D: Debt or borrowings from financial institutions

r_d : Cost of debt funding (interest rate)

Balance sheet



Funds
investment

Funds
procurement

Total Project Cost	100	M\$	(E+D)
Loan amount	70	M\$	(D)
Equity amount	30	M\$	(E)
Interest rate	6%		(r_d)
Cost of Equity	12%		(r_e)
Loan portion	70%		(D)/(E+D)
Equity portion	30%		(E)/(E+D)
WACC	7.80%		Loan portion * r_d + Equity portion * r_e

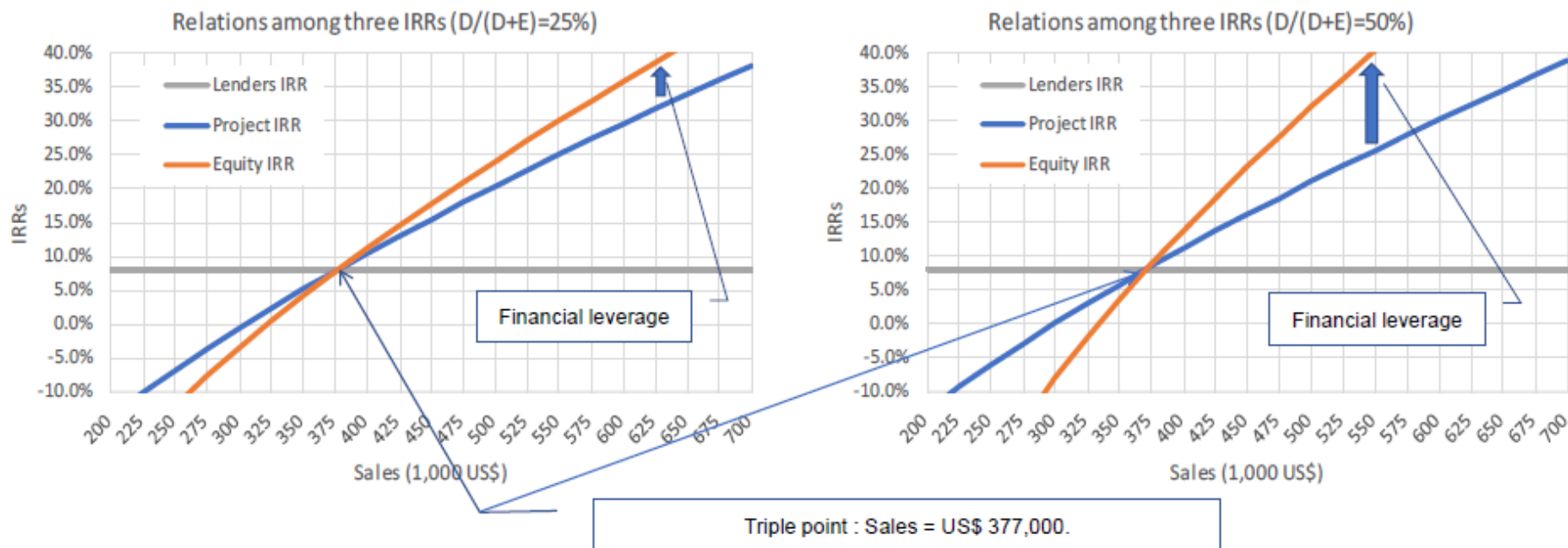
WACC is 7.80%

Relations among three IRRs

Let's look at IRRs of three parts. We can see,

- Lender's IRR is constant (always equals to interest rate).
- If Project IRR > Interest rate, Equity IRR > Project IRR.
- If Project IRR < Interest rate, Equity IRR < Project IRR.

The phenomenon that **Equity IRR > Project IRR** is called “**Leverage effect of borrowings**”.
The leveraged effect becomes stronger when the borrowing ratio ($D/(D+E)$) becomes larger.



[Example 4.8] (PrIRR & EqIRR)

A simple power plant business

[Conditions]

This is a power plant project with **50 MW**.

Operation period is **5 years**.

Annual sales volume is **370 GWh**.

Sales price of electricity is **US 10 ct/kWh**.

Annual sales amount is **US\$ 37 million**.

Construction cost is **US\$ 100 million**. Construction is done in one year.

Self-fund is US\$ 40 million and borrowing is US\$ 60 million.

Repayment period is **5 years**.

Interest rate is **6%**.

Repayment method is **equal-repayment-of-principle**.

Don't consider depreciation for simplicity purpose.

Annual O&M cost is **US\$ 2 million**.

Tax rate is **30%**.

[Example 4.8] A simple power plant business

Calculate Project IRR and Equity IRR with Excel, Sheet Q4.8.

	A	B		D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
28																		
29																		
30																		
31																		
32																		
33																		
34																		
35																		
36																		
37																		
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72																		
73																		
74																		
75																		
76																		
77																		

Q1 Create a five-year income statement for this project between Row-51 and Row-57.

Q2 Calculate free cash flow to the project (FCFP) and Project IRR between Row-61 and Row-63.

Q3 Calculate free cash flow to equity (FCFE) and Equity IRR between Row-69 and Row-73.

[Tips] Sales amount
 $370 \text{ GWh} \times 10 \text{ ct/kWh} = 370 \times 10^6 \text{ kWh} \times 0.10 \text{ \$/kWh}$
 $= 37 \times 10^6 \text{ \$}$
 $= 37 \text{ M\$}$

Answer

Displayed formula

Please fill in70

A Simple Financial Model for a Total Geothermal Project

[Example 5] A total geothermal project

[Conditions]

This is a geothermal project with **70 MW**. Operation period is **25 years**. Annual sales volume is **500 GWh**. Sales price of electricity is **US 10 ct/kWh**. Annual sales amount is **US\$ 50 million**. Construction cost is **US\$ 300 million**. Construction is done in 5 years.

Cost item	M\$	Construction schedule					
		Year-(-5)	Year-(-4)	Year-(-3)	Year-(-2)	Year-(-1)	Total
1. Reconnaissance	5	5					5
2. Exploration	35	15	20				35
3. Construction (Drilling)	60		30	30			60
4. Construction (FCRS)	50			20	20	10	50
5. Power Plant	150			50	50	50	150
TOTAL	300	20	50	100	70	60	300
Equity	140	20	50	30	20	20	140
Debt	160			70	50	40	160

Total self-fund (equity) is **US\$ 140 million** and borrowing is **US\$ 160 million**.

Repayment period is **8 years**.

Interest rate is **6%**.

Repayment method is **equal-repayment-of-principle**.

Depreciation is **straight line method of 15 years**.

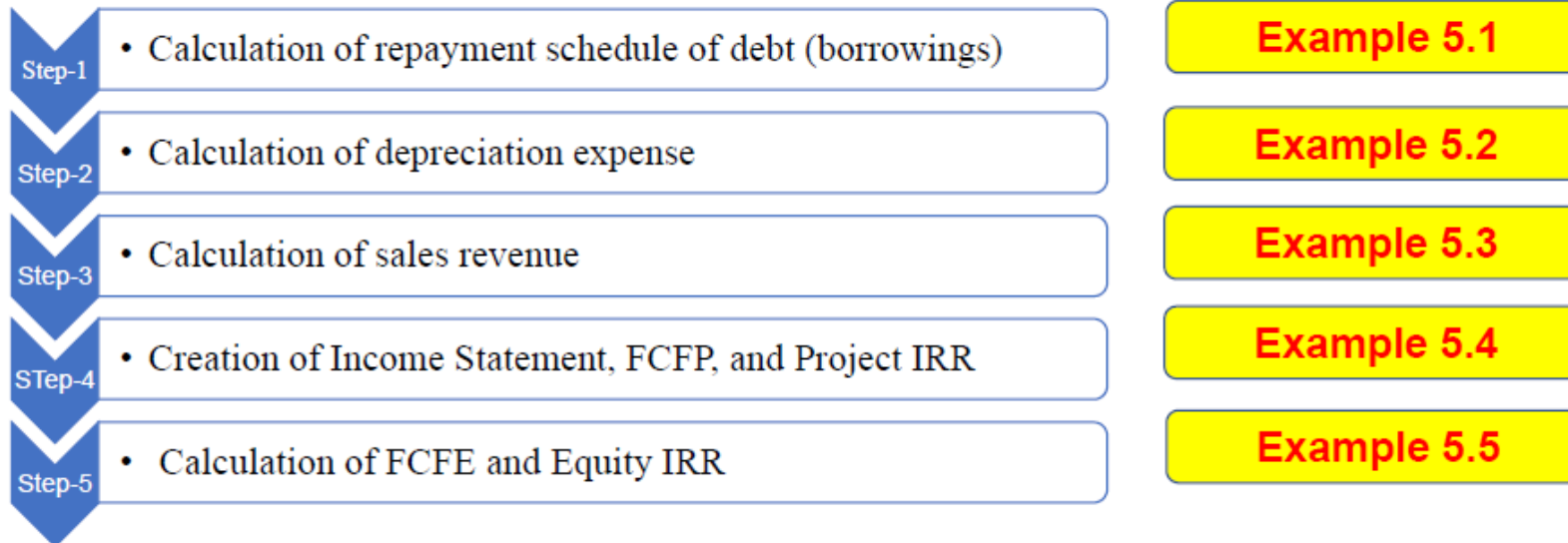
Annual wage is **US\$ 2 million** and annual O&M cost is **US\$ 5 million**.

Tax rate is **30%**.

[Example 5] A total geothermal project

Let's create a financial model with Excel, Sheet Q5.

Please follow the steps shown below.



[Example 5.1] Calculation of Repayment Plan

Let's create a repayment schedule with Excel, Sheet Q5.

Project cost & Construction Schedule										
Item	Unit	5	4	3	2	1	Total			
1. Reconnaissance	HS	5					5			
2. Exploration	HS	15	30				45			
3. Construction (Lifting)	HS	10	30	30			70			
4. Construction (FORS)	HS	10		20	20	10	50			
5. Power Plant	HS	110		50	50	50	260			
TOTAL	HS	300	20	110	100	70	580			
STEP 4.2. Cash Flow & Debt schedule										
Item	Unit	5	4	3	2	1	Total			
CAPEX expenditure	HS	20	60	100	70	50	300			
Isolity	HS	20	60	20	20	20	140			
Borrowings	HS			70	50	40	160			
STEP 2.1. Calculation of IDC										
Item	Unit	5	4	3	2	1	Total			
Borrowing in Year (5)	HS	70	70.2	70.7	71.2	71.7	353.8			
Borrowing in Year (4)	HS		50	50.0	50.0	50.0	150.0			
Borrowing in Year (3)	HS			40	40.0	40.0	120.0			
Borrowing with IDC Total	HS			70.6	134.9	171.7	377.2			
IDC (nom.)	HS			4.2	11.7	11.7	27.6			
IDC actually accrued	HS			4.2	7.5	-	11.7			
STEP 2.2. Calculation of repayment schedule										
Item	Unit	5	4	3	2	1	Total			
Principal repayment	HS									
Interest payment	HS									
Total repayment	HS									
Balance of the year end	HS									
Lender's Cash Flow	HS									
Lender's IRR	%									
STEP 3. Depreciation cost										
Item	Unit	5	4	3	2	1	Total			
Depreciation	HS									
Residual value (year=5)	HS									
STEP 4. Annual value										
Item	Unit	5	4	3	2	1	Total			
Crash sales amount	HS									
CRASH	HS									

Q1 Make a repayment plan of principal in Row 60.

Q2 Calculate an outstanding balance of the borrowing at the end of each year in Row 63.

Q3 Make an interest payment plan of the borrowing in Row 61.

Q4 Calculate a total repayment in Row 62.

(*) Calculate Lender's IRR from the obtained cash flow. Check it whether the Lender's IRR is equal to the interest rate (6%).

[Example 5.1] Calculation of Repayment Plan

Answer

Project cost & Construction Schedule												
Item	M\$	Unit	-5	-4	-3	-2	-1	Total				
1. Reconnaissance	5	M\$	5					5				
2. Exploration	35	M\$	15	20				35				
3. Construction (Drilling)	60	M\$		30	30			60				
4. Construction (FCRS)	50	M\$			20	20	10	50				
5. Power Plant	150	M\$			50	50	50	150				
TOTAL	300	M\$	20	50	100	70	60	300				

Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13
STEP.1 CAPEX, Equity & Debt schedule																			
CAPEX expenditure	(0)	M\$	20	50	100	70	60												
Equity	(0)	M\$	20	50	30	20	20												
Borrowings	(0)	M\$			70	50	40												
STEP.2.1 Calculation of IDC																			
Borrowing in Year (-3)	M\$				70	74.2	78.7												
Borrowing in Year (-2)	M\$					50	53.0												
Borrowing in Year (-1)	M\$						40												
Borrowing with IDC Tr					70.0	124.2	171.7	171.7											
IDC (cum.)						4.2	11.7	11.7											
IDC annually accrued						4.2	7.5	-											
STEP.2.2 Calculation of repayment schedule																			
Item	Unit	Total	1	2	3	4	5	6	7	8	9	10	11	12	13				
Principal repayment	(33)	M\$	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5									
Interest payment	(32)	M\$	10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3									
Total repayment	(33)+(32)	M\$	31.8	30.5	29.2	27.9	26.6	25.3	24.0	22.7									
Balance of the year end	(31)	M\$	171.7	150.2	128.7	107.3	85.8	64.4	42.9	21.5	-0.0	-0.0	-0.0	-0.0	-0.0				
Lender's Cash Flow	M\$		-70.0	-50.0	-40.0	31.8	30.5	29.2	27.9	26.6	25.3	24.0	22.7						
Lender's IRR	6.00%	OK																	

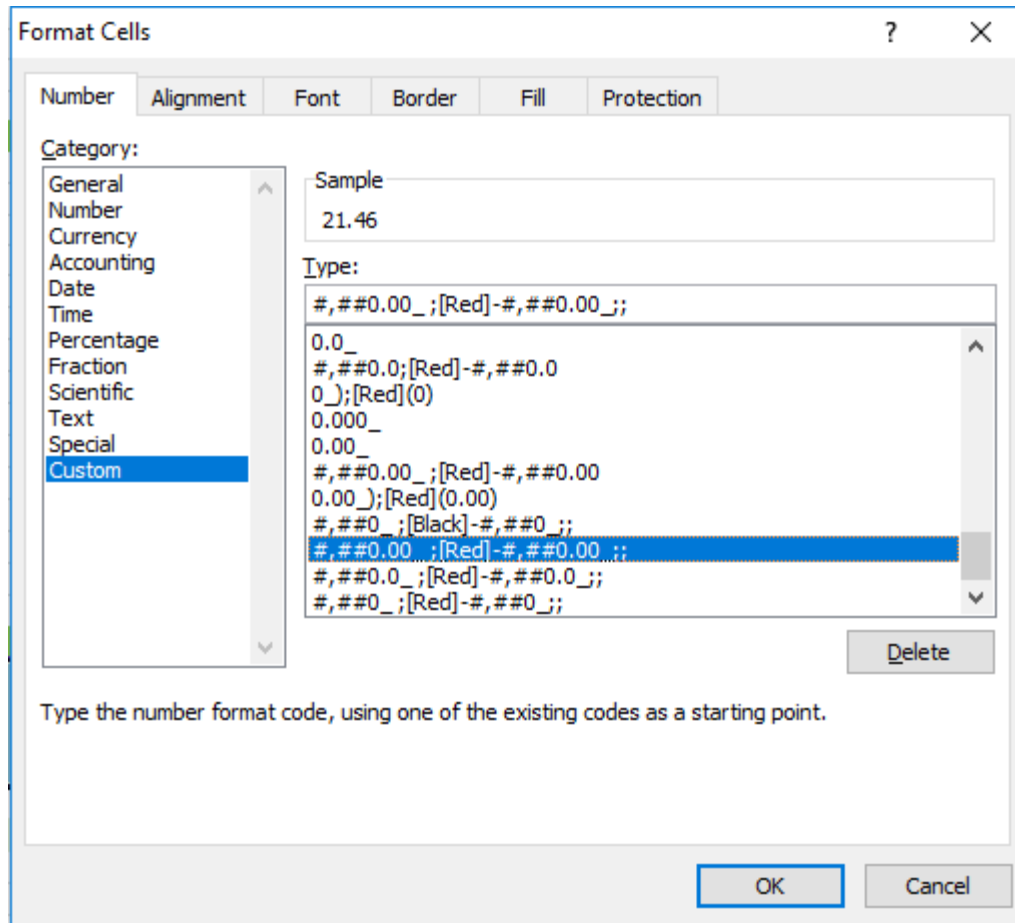
Principal repayment
=IF(M59<=\$E\$17,\$L\$63/\$E\$17,0)

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Interest payment
=IF(M59<=\$E\$17,L63*\$E\$18,0)

Balance of borrowing
=L63-M60

Cell Format



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The format is set:

[cell format setting] → [user definition]
→ [# , ## 0.00 _; [red] - #, ## 0.00 _ ;;]

The meaning;

- Positive number 12,345.67
- Negative number -12,345.67
- **Zero** **Blank (not shown).**

[Example 5.2] Calculation of Depreciation expense

Let's calculate depreciation expense with Excel, Sheet Q5.

Project cost & Construction Schedule									
Item	Unit	-5	-4	-3	-2	-1	Total		
1. Reconnaissance	MS	5					5		
2. Exploration	MS	25	15	20			60		
3. Construction (Drilling)	MS	80		30			110		
4. Construction (FORS)	MS	30			20	20	70		
5. Power Plant	MS	150			10	10	170		
TOTAL	MS	300	20	50	30	30	430		
STEP 1 CAPEX Equity & Debt schedule									
Item	Unit	-5	-4	-3	-2	-1	1	2	3
CAPEX expenditure	MS	20	50	100	30	30			
Equity	MS	20	50	30	20	20			
Borrowings	MS			70	10	10			
STEP 2.1 Calculation of IDC									
Item	Unit	-5	-4	-3	-2	-1	Total		
Borrowing in Year (-5)	MS						70.7		
Borrowing in Year (-2)	MS				50	50.0	50.0		
Borrowing in Year (-1)	MS					40	40.0		
Borrowing with IDC Total	MS				70.7	124.2	174.7		
IDC (sum)	MS				4.2	11.7	11.7		
IDC annually accrued	MS				4.2	7.5	-		
STEP 2.2 Calculation of repayment schedule									
Item	Unit						Total	1	2
Principal repayment	MS								
Interest payment	MS								
Total repayment	MS								
Balance of the year end	MS						174.7		
Lender's Cash Flow	MS						-70.7	-50.0	-40.0
Lender's IRR	%						10.00%		
STEP 3 Depreciation cost									
Item	Unit						Total	1	2
Depreciation	MS								
Book value (year-end)	MS						214.7		
STEP 4 Annual sales									
Item	Unit	-5	-4	-3	-2	-1	1	2	3
Energy sales amount	(Y=CellE7)								
Sales	MS								

Q1 Input the calculation formula of the annual depreciation expense of the plant into Row 70.

Q2 Input the calculation formula of the book value of the plant at the end of the year into Row 71.

[Example 5.2] Calculation of Depreciation expense

Answer

STEP-2.2 Calculation of repayment schedule				Total	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Item		Unit																			
Principal repayment	(33)	M\$			21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5									
Interest payment	(32)	M\$			10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3									
Total repayment	(33)+(32)	M\$			31.8	30.5	29.2	27.9	26.6	25.3	24.0	22.7									
Balance of the year end	(31)	M\$			171.7	150.2	128.7	107.3	85.8	64.4	42.9	21.5	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Lender's Cash Flow		M\$		-70.0	-50.0	-40.0	31.8	30.5	29.2	27.9	26.6	25.3	24.0	22.7							
Lender's IRR	6.00%	OK																			
STEP-3 Depreciation cost				Total	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Item		Unit																			
Depreciation	(2)=TTL Cost/Dep Yr	M\$		20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8
Book value (year-end)	(22)	M\$		311.7	290.9	270.1	249.3	228.5	207.8	187.0	166.2	145.4	124.7	103.9	83.1	62.3	41.6	20.8			

Depreciation expense
=IF(M69<=\$E\$26,\$L\$71/\$E\$26,0)

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Book value
=IF(M69<=\$E\$26,L71-M70,0)

[Example 5.3] Calculation of Sales Revenue

Let's create a financial model with Excel, Sheet Q5.

Q1 Create sales volume (GWh) for this project in Row-75.

Q2 Calculate sales revenue (M\$) in Row-76.

72																									
73	STEP-4 Annual sales																								
74	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
75	Energy sales amount	(1)=Cell(E7)	GWh																						
76	Sales	(1)=Cell(E8)	MS																						
77																									
78	STEP-5 Income Statement (Profit & Loss table)																								
79	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
80	Sales	(1)	MS																						
81	Wage	(51)=Cell(E24)	MS																						
82	Operation & Maintenance cost	(52)=Cell(E25)	MS																						
83	Depreciation expense	(2)	MS																						
84	Earning before Interest & Tax (EBIT)	(6)=(1)-(51)-(52)-(2)	MS																						
85	Interest payment	(32)	MS																						
86	Income before Tax	(7)=(6)-(32)	MS																						
87	Tax	(8)=(7)*tax rate	MS																						
88	Income after tax	(9)=(7)-(8)	MS																						
89																									

Answer

Note: Price is in cents terms. Need to divide by 100 to convert it into dollar terms.

[Tips] Sales amount
 $500 \text{ GWh} \times 10 \text{ ct/kWh} = 500 \times 10^6 \text{ kWh} \times 0.10 \text{ \$/kWh}$
 $= 50 \times 10^6 \text{ \$}$
 $= 50 \text{ M\$}$

[Example 5.4] Calculation of Income Statement, FCFP, and Project IRR

Let's make Income Statement, FCFP, Project IRR in Sheet Q5.

73	STEP 4 Annual sales																								
74	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		
75	Energy sales amount	(1)=Cell(E1)	GW/h																						
76	Sales	(1)=Cell(E1)	US																						
77																									
78	STEP 5 Income Statement (Profit & Loss table)																								
79	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		
80	Sales	(1)	US																						
81	Wage	(51)=Cell(E24)	US																						
82	Operation & Maintenance cost	(52)=Cell(E25)	US																						
83	Depreciation expense	(2)	US																						
84	Earning before Interest & Tax (EBIT)	(6)=(1)-(51)-(52)-(2)	US																						
85	Interest payment	(32)	US																						
86	Income before Tax	(7)=(6)-(32)	US																						
87	Tax	(8)=(7)*tax rate	US																						
88	Income after tax	(9)=(7)-(8)	US																						
89																									
90	STEP 6 Free Cash Flow (FCF) and Project IRR (PIRR)																								
91	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		
92	Income after tax	(9)	US																						
93	Depreciation expense	(2)	US																						
94	Interest payment	(32)	US																						
95	Free Cash Flows to Project (FCFP)	(10)=(9)-(2)-(32)	US	-20.0	-50.0	-100.0	-70.0	-40.0																	
96	Project IRR (PIRR)=																								
97	WACC=	8.81%																							
98																									
99	STEP 7 Free Cash Flow to Equity and Equity IRR (EqIRR)																								
100	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		
101	Free Cash Flows to Project (FCFP)	(10)	US	-20.0	-50.0	-100.0	-70.0	-40.0																	
102	Borrowings	(07)	US				70.0	50.0	40.0																
103	Interest payment	(32)	US																						
104	Principle repayment	(33)	US																						
105	FCF to Equity	(12)=(10)-(07)-(32)-(33)	US	-20.0	-50.0	-30.0	-20.0	-20.0																	
106	Equity IRR (EqIRR)=																								
107	Expected Return on Equity (ROE)=	12.01%																							

Q1 Make an Income Statement in the rows between Row-80 and Row-88.

Q2 Calculate free cash flow to the project (FCFP) using the rows between Row-91 and Row-96.

Q3 Calculate Project IRR in Cell (96E) using Row-95.

[Example 5.4] Calculation of Income Statement, FCFP, and Project IRR

Answer

STEP-5 Income Statement (Profit & Loss table)																			
Item		Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12
Sales	(1)	M\$						50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0
Wage	(51)=Cell(E24)	M\$						2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Operation & Maintenance cost	(52)=Cell(E25)	M\$						5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Depreciation expense	(2)	M\$						20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8
Earning before Interest & Tax (EBIT)	(6)=(1)-(51)-(52)-(2)	M\$						22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2
Interest payment	(32)	M\$						10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3	-0.0	-0.0	-0.0	-0.0
Income before Tax	(7)=(6)-(32)	M\$						11.9	13.2	14.5	15.8	17.1	18.4	19.6	20.9	22.2	22.2	22.2	22.2
Tax	(8)=(7)*tax rate	M\$						3.0	3.3	3.6	3.9	4.3	4.6	4.9	5.2	5.6	5.6	5.6	5.6
Income after tax	(9)=(7)-(8)	M\$						8.9	9.9	10.9	11.8	12.8	13.8	14.7	15.7	16.7	16.7	16.7	16.7
STEP-6 Free Cash Flow (FCF) and Project IRR (PrIRR)																			
Item		Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12
Income after tax	(9)	M\$						8.9	9.9	10.9	11.8	12.8	13.8	14.7	15.7	16.7	16.7	16.7	16.7
Depreciation expense	(2)	M\$						20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8
Interest payment	(32)	M\$						10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3	-0.0	-0.0	-0.0	-0.0
Free Cash Flows to Project (FCFP)	(10)=(9)+(2)+(32)	M\$						40.0	39.7	39.4	39.1	38.7	38.4	38.1	37.8	37.4	37.4	37.4	37.4
Project IRR (PrIRR)=		9.59%																	
WACC=		8.80%																	

Project IRR = 9.59%

[Example 5.5]

Calculation of FCFE and Equity IRR

Let's calculate FCFE and Equity IRR in Sheet Q5.

73	STEP 4 Annual sales																							
74	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
75	Energy sales amount	(1)=C6(E7)	GWh																					
76	Sales	(1)=C6(E8)	M\$																					
77																								
78	STEP 5 Income Statement (Profit & Loss table)																							
79	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
80	Sales	(1)	M\$																					
81	Wage	(51)=C6(E24)	M\$																					
82	Operation & Maintenance cost	(52)=C6(E25)	M\$																					
83	Depreciation expense	(2)	M\$																					
84	Earning before Interest & Tax (EBIT)	(6)=(1)-(51)-(52)	M\$																					
85	Interest payment	(32)	M\$																					
86	Income before Tax	(7)=(6)-(32)	M\$																					
87	Tax	(8)=(7)*tax rate	M\$																					
88	Income after tax	(9)=(7)-(8)	M\$																					
89																								
90	STEP 6 Free Cash Flow (FCF) and Project IRR (PIRR)																							
91	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
92	Income after tax	(9)	M\$																					
93	Depreciation expense	(2)	M\$																					
94	Interest payment	(32)	M\$																					
95	Free Cash Flows to Project (FCFP)	(10)=(9)+(2)-(32)	M\$		-20.0	-50.0	-100.0	-70.0	-60.0															
96	Project IRR (PIRR)=																							
97	WACC=	8.00%																						
98																								
99	STEP 7 Free Cash Flow to Equity and Equity IRR (EqIRR)																							
100	Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
101	Free Cash Flows to Project (FCFP)	(10)	M\$		-20.0	-50.0	-100.0	-70.0	-60.0															
102	Borrowings	(01)	M\$				70.0	50.0	40.0															
103	Interest payment	(32)	M\$																					
104	Principle repayment	(33)	M\$																					
105	FCF to Equity	(12)=(10)+(01)-(32)-(33)	M\$		-20.0	-50.0	-30.0	-20.0	-20.0															
106	Equity IRR (EqIRR)=																							
107	Expected Return on Equity (ROE)=	12.00%																						
108																								

Q1 Create free cash flow to equity (FCFE) in the rows between Row-101 and Row-106.

Q2 Calculate Equity IRR in Cell (106E) using FCFE (Row-105).

[Example 5.5]

Calculation of FCFE and Equity IRR

Answer

STEP-5 Income Statement (Profit & Loss table)																	
Item		Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10
Sales	(1)	MS						50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0
Wage	(51)=Cell(E24)	MS						2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Operation & Maintenance cost	(52)=Cell(E25)	MS						5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Depreciation expense	(2)	MS						20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8
Earning before Interest & Tax (EBIT)	(6)=(1)-(51)-(52)-(2)	MS						22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2
Interest payment	(32)	MS						10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3	-0.0	-0.0
Income before Tax	(7)=(6)-(32)	MS						11.9	13.2	14.5	15.8	17.1	18.4	19.6	20.9	22.2	22.2
Tax	(8)=(7)*tax rate	MS						3.0	3.3	3.6	3.9	4.3	4.6	4.9	5.2	5.6	5.6
Income after tax	(9)=(7)-(8)	MS						8.9	9.9	10.9	11.8	12.8	13.8	14.7	15.7	16.7	16.7
STEP-6 Free Cash Flow (FCF) and Project IRR (PrIRR)																	
Item		Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10
Income after tax	(9)	MS						8.9	9.9	10.9	11.8	12.8	13.8	14.7	15.7	16.7	16.7
Depreciation expense	(2)	MS						20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8	20.8
Interest payment	(32)	MS						10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3	-0.0	-0.0
Free Cash Flows to Project (FCFP)	(10)=(9)+(2)-(32)	MS	-20.0	-50.0	-100.0	-70.0	-60.0	40.0	39.7	39.4	39.1	38.7	38.4	38.1	37.8	37.4	37.4
Project IRR (PrIRR)=	9.59%																
WACC=	8.80%																
STEP-7 Free Cash Flow to Equity and Equity IRR (EqIRR)																	
Item		Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10
Free Cash Flows to Project (FCFP)	(10)	MS	-20.0	-50.0	-100.0	-70.0	-60.0	40.0	39.7	39.4	39.1	38.7	38.4	38.1	37.8	37.4	37.4
Borrowings	(0)	MS	0.0	0.0	70.0	50.0	40.0										
Interest payment	(32)	MS						10.3	9.0	7.7	6.4	5.1	3.9	2.6	1.3	-0.0	-0.0
Principle repayment	(33)	MS						21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5
FCF to Equity	(12)=(10)+(0)-(32)-(33)	MS	-20.0	-50.0	-30.0	-20.0	-20.0	8.3	9.2	10.2	11.2	12.1	13.1	14.1	15.0	15.9	16.9
Equity IRR (EqIRR)=	10.79%																
Expected Return on Equity (ROE)=	12.00%																

Equity IRR is 10.79%.

[Example 5.6]

Check of created Financial Model

Let's check your model in Sheet Q5.

Try to find the sales price that realizes the following equation in your model.

Project IRR = Equity IRR = Lender's IRR (= 6%)

[Tips]

You may use the goal seek function of Excel, or you may change sales price (Cell E8) by manual on a trial and error basis.

[Tips] Examination of Triple Point

The point that makes 'Project IRR = Equity IRR = Lenders IRR' is called 'triple point.'

If your model does not have the triple point, the model may have some mistakes. In such a case, you need to check the model.

If your model has the triple point, your model is likely to be correctly created.

Answer

[Example 5.6]

Check of created Financial Model

Let's check your model in Sheet Q5.

Try to find the sales price that realizes the following equation in your model.

Project IRR = Equity IRR = Lender's IRR (= 6%)

[Tips]

You may use the goal seek function of Excel, or you may change sales price (Cell E8) by manual on a trial and error basis.

[Tips] Examination of Triple Point

The point that makes 'Project IRR = Equity IRR = Lenders IRR' is called 'triple point.'

If your model does not have the triple point, the model may have some mistakes. In such a case, you need to check the model.

If your model has the triple point, your model is likely to be correctly created.

Answer

When the sales price is around **7.08 ct/kWh**, the relation appears.

Project IRR = Equity IRR = Lender's IRR (= 6.00%)

[Example 5.7]

Sensitivity Analysis

Let's use your model in Sheet Q5.

Q1 Q Calculate Project IRR, Equity IRR and Lender's IRR when the sales price is 6, 8, 10, 12, 14 and 16 ct/kWh.

Sales price (ct/kWh)	Project IRR	Equity IRR	Lender's IRR
6.0			
8.0			
10.0			
12.0			
14.0			
16.0			

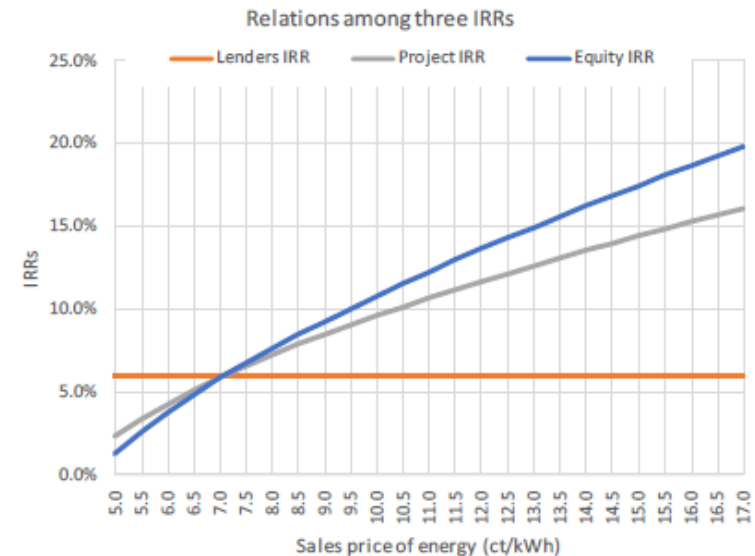
[Example 5.7]

Sensitivity Analysis

Answer

Sales price (ct/kWh)	Project IRR	Equity IRR	Lender's IRR
6.0	4.3%	3.8%	6.0%
8.0	7.2%	7.6%	6.0%
10.0	9.6%	10.8%	6.0%
12.0	11.7%	13.6%	6.0%
14.0	13.5%	16.2%	6.0%
16.0	15.3%	18.6%	6.0%

Sensitivity analysis of sales price on IRRs



[Example 5.8]

Lending condition of bank

Let's use your model in Sheet Q5.

- Q Which bank's offer is the most advantageous to your project?
Calculate Equity IRR at the sales price of 10 ct/kWh.

Bank	Lending condition		Profitability
	Interest rate	Repayment period	Equity IRR
Bank-A	5.0%	7 years	
Bank-B	6.0%	8 years	10.79%
Bank-C	7.0%	9 years	

[Example 5.8]

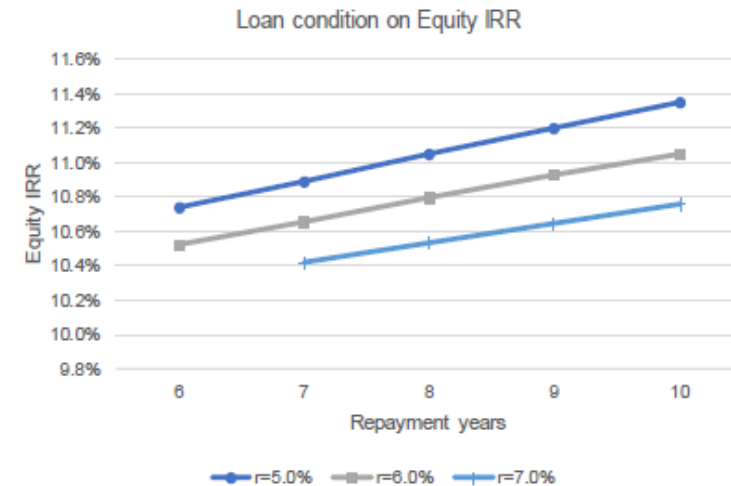
Lending condition of bank

Answer

The offers of three banks are almost equivalent in terms of Equity IRR of this project.
But Bank A's offer is the most advantageous slightly.

Bank	Lending condition		Profitability
	Interest rate	Repayment period	Equity IRR
Bank-A	5.0%	7 years	10.89%
Bank-B	6.0%	8 years	10.79%
Bank-C	7.0%	9 years	10.65%

Sensitivity analysis of loan condition on Equity IRR



[Example 5.9]

Application of Financial Model

Let's use your model in Sheet Q5.

Q Seek for the minimum sales price which achieves 10%, 12%, 15% and 20% of Equity IRR.

Equity IRR	10%	12%	15%	20%
Sales price (ct/kWh)				

[Tips]

Use “Goal Seek” function

Data ribbon → What-If analysis

→ Goal seek

Item	MS	Unit	-5	-4	-3	-2	-1	Total
1. Reconnaissance	5	MS						5
2. Exploration	15	MS	15					15
3. Construction (Drilling)	80	MS		30	30			80
4. Construction (PCRS)	100	MS			20	20	70	110
5. Power Plant	100	MS				40	60	140
TOTAL	200	MS	15	30	50	70	60	200

Item	Unit	-5	-4	-3	-2	-1	1	2	3	4	5	6	7
STEP 1 CAPEX, Equity & Debt schedule													
CAPEX expenditure	(M)	MS	20	50	100	70	60						
Equity	(M)	MS	20	50	20	20	20						
Borrowings	(M)	MS											

Set cell (E): Cell of Equity IRR : [\$32\$E]

To value : Target = 10%

By changing cell (C): Cell of sales price : [\$8\$E]

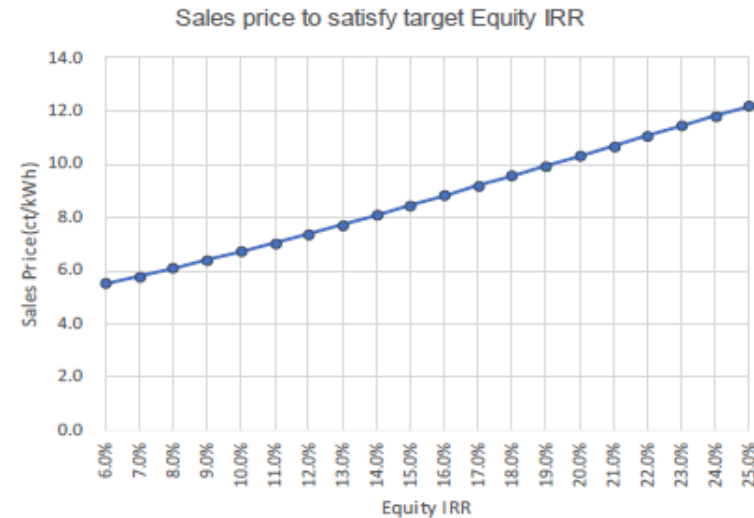
[Example 5.9]

Application of Financial Model

Answer

Equity IRR	10%	12%	15%	20%
Sales price (ct/kWh)	9.46	10.82	13.03	17.19

Since the capital cost accounts for most of geothermal energy price, the expected return of the project operator (Equity IRR) affects the energy price largely as shown in the above Table or the right Figure.



Discounted Profitability Index (DPI)

Discounted Profitability Index (DPI)

Profitability Index is a ratio of discounted cash inflow to the discounted cash outflow. Discounted cash inflow is our benefit in the project and the initial investment is our cost, which is why we also call it benefit to cost ratio.

$$DPI = \frac{NPV \text{ cash inflow}}{NPV \text{ initial investment}}$$

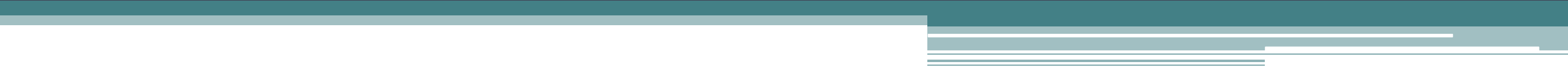
DPI dapat digunakan sebagai paramater acuan untuk memutuskan akan dilakukan/tidak dilakukan sebuah proyek/*initial investment*. Batas Nilai DPI minimal bergantung pada *company* masing-masing, misalnya >1.2 atau >1.3

Conclusion

Project feasibility / investment judgement criteria:

- $NPV > 0$
- Project IRR > Hurdle rate or WACC
- $DPI \geq 1.2$ or 1.3

Thank you

A series of horizontal lines in teal and light blue colors, with varying lengths and offsets, creating a modern, layered effect across the bottom of the slide.